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Multi-Agent Governance for Graph-Regularized Conditional Value-at-Risk Portfolio Optimization with Adaptive Contagion Penalization

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Abstract:	One of the most significant problems for institutional portfolio managers during financial crises is that contagion can spread across sectors via forced selling of institutional holdings that overlap. Classic Conditional Value-at-Risk (CVaR) models are designed to measure tail risk but do not account for common institutional ownership spillovers. We propose a multi-agent governance framework for graph-regularized Conditional Value-at-Risk portfolio optimization with adaptive contagion penalization. The framework adopts a five-agent blackboard architecture for data ingestion, instability analysis, contagion graph construction, portfolio optimization, explainability and governance auditing. We model institutional co-ownership relationships with a bipartite holdings graph, where highly connected securities are adaptively penalized based on eigenvector centrality, and a sigmoid-gated instability mechanism. The framework is tested on 11 Global Industry Classification Standard sector universes, 218 United States equities and 552 rolling windows over the 2005-2025 period, which includes the 2008 Global Financial Crisis, the 2020 COVID shock and the 2022 rate-hike episode. The results show a 25.9% reduction in Conditional Value-at-Risk at the 95% confidence level and a 32.5 percentage point reduction in crisis period drawdown relative to equal weight benchmarks. Furthermore, crisis-window analysis also shows statistically significant improvements in tail-risk containment. The proposed architecture also offers fault isolation, auditability, replayability and human-in-the-loop governance for regulatory-compliant portfolio decision making.

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Cover letter to the Editor

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of the journal of Engineering Applications of Artificial Intelligence

Dear Editors,

We are submitting our original research article entitled, “Multi-Agent Governance for Graph-Regularized Conditional Value-at-Risk Portfolio Optimization with Adaptive Contagion Penalization” for consideration in *Engineering Applications of Artificial Intelligence*. We confirm that the manuscript is original and is not under consideration in another journal.

This work presents a multi-agent governance framework for portfolio optimization under financial contagion. The system combines graph-based risk modeling, adaptive penalization, human-in-the-loop oversight, and blackboard-based coordination to support transparent and auditable financial decision-making.

The manuscript evaluates the framework across 11 sector universes and 552 rolling windows from 2005 to 2025, including major financial crisis periods. The proposed graph-regularized Conditional Value-at-Risk optimization reduces average tail risk by 25.9% and reduces crisis-period maximum drawdown by 32.5 percentage points relative to equal-weight benchmarks.

We believe the manuscript is well aligned with the scope of *Engineering Applications of Artificial Intelligence* because it demonstrates the application of intelligent multi-agent systems to real-world financial risk governance and portfolio optimization.

We have no conflicts of interest to disclose.

Thank you for your time and consideration.

Sincerely,

Kandula Jithendra Subramanyam

(On behalf of all authors)

Research Highlights

- Multi-agent governance framework for contagion-aware portfolio optimization.
- Graph regularization penalizes highly connected institutional holdings.
- The framework reduces tail risk during major financial crisis periods.
- Human-in-the-loop governance improves auditability and fault isolation.
- Results remain robust across 11 sector universes and 552 rolling windows.

Multi-Agent Governance for Graph-Regularized Conditional Value-at-Risk Portfolio Optimization with Adaptive Contagion Penalization

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CRediT Authorship Contribution Statement

Kandula Jithendra Subramanyam: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – original draft, Writing – review and editing, Visualization, Investigation.

Sunayana Vilas Jadhav: Supervision, Writing – review and editing.

Dr. Ashwini Dalvi: Supervision, Writing – review and editing.

Declaration of Competing Interests

The authors declare no known competing financial interests or personal relationships that could have influenced this work.

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Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests

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ABSTRACT

One of the most significant problems for institutional portfolio managers during financial crises is that contagion can spread across sectors via forced selling of institutional holdings that overlap. Classic Conditional Value-at-Risk (CVaR) models are designed to measure tail risk but do not account for common institutional ownership spillovers.

We propose a multi-agent governance framework for graph-regularized Conditional Value-at-Risk portfolio optimization with adaptive contagion penalization. The framework adopts a five-agent blackboard architecture for data ingestion, instability analysis, contagion graph construction, portfolio optimization, explainability and governance auditing. We model institutional co-ownership relationships with a bipartite holdings graph, where highly connected securities are adaptively penalized based on eigenvector centrality, and a sigmoid-gated instability mechanism.

The framework is tested on 11 Global Industry Classification Standard sector universes, 218 United States equities and 552 rolling windows over the 2005–2025 period, which includes the 2008 Global Financial Crisis, the 2020 COVID shock and the 2022 rate-hike episode. The results show a 25.9% reduction in Conditional Value-at-Risk at the 95% confidence level and a 32.5 percentage point reduction in crisis period drawdown relative to equal weight benchmarks. Furthermore, crisis-window analysis also shows statistically significant improvements in tail-risk containment. The proposed architecture also offers fault isolation, auditability, replayability and human-in-the-loop governance for regulatory-compliant portfolio decision making.

1. Introduction

1.1. The Co-Ownership Problem in Institutional Portfolios

Overlapping ownership is a quiet feature of normal markets. It becomes a dangerous one when stress arrives. In 2008, the forced selling that began in leveraged credit positions rapidly spread across equity sectors that had no obvious fundamental connection — connected instead by the fact that the same large institutions held all of them and needed to sell them simultaneously (Allen and Gale, 2000; Billio, Getmansky, Lo and Pelizzon, 2012). In addition to this, there is one more factor which contributed to the 2020 experience, and that was the situation of deleveraging, which took place in different industries. This was unexpected, as these industries showed no prior evidence of correlated stress, a channel that conventional risk models failed to anticipate. In the 2022 rate-hike episode, rising interest rates generated systemic risk through losses borne by asset-management firms with high yield and duration exposures.

They happen again and again because, in conventional portfolio optimisation, the co-ownership effect is not incorporated into the analysis. Mean-variance portfolio optimisation and Conditional Value-at-Risk (CVaR) both consider risk to be a characteristic of the probability distribution of the returns on each financial instrument separately (Markowitz, 1952; Rockafellar and Uryasev, 2000). However, they ignore the reality that a stock can appear in multiple large institutional portfolios simultaneously, and that such portfolios can try to exit the market en masse within the same time window during times of turmoil. Thus, a portfolio might be considered well-diversified using conventional metrics while being fragile from the perspective of co-ownership.

There is also a governance problem. The European Union Artificial Intelligence Act (EU AI Act) (European Parliament and Council of the EU, 2024) and MiFID II (European Parliament and Council of the EU, 2014) require investment systems to leave a traceable, human-readable record of important rebalancing decisions. A black-box optimiser, no matter how accurate, does not meet that standard on its own. The approach in this paper is meant to address both issues together.

1.2. Research Questions and Hypotheses

Data-driven portfolio methods, including reinforcement learning (Jiang, Xu and Liang, 2017), transformer-based return prediction (Gu, Kelly and Xiu, 2020), and financial LLMs such as BloombergGPT (Wu et al., 2023), have

reported strong in-sample results. In stressed markets, though, they can be brittle, susceptible to numerical hallucination (Ji et al., 2023), and difficult to audit in a regulatory setting (Arrieta et al., 2020). More traditional methods such as convex optimisation (Boyd and Vandenberghe, 2004; Kolm, Tütüncü and Fabozzi, 2014), regime models (Hamilton, 1989), and graph contagion analysis (Billio et al., 2012) are easier to reproduce, but they are also more static and do not directly address co-ownership effects. The multi-agent approach using the blackboard paradigm (Hayes-Roth (1985); Erman, Hayes-Roth, Lesser and Reddy (1980)) is an effective alternative between the two ends of the spectrum.

RQ1. Would co-ownership structure of institutions, represented in a two-mode network of assets versus institutions, be able to create a meaningful signal in penalizing concentration?

H1. High eigenvector centrality assets in the investment graph suffer much higher losses during a systemic shock, and thus merit punishment within the CVaR criterion.

RQ2. Does adaptive sigmoid-gated graph penalisation deliver measurable tail-risk improvements across 11 Global Industry Classification Standard (GICS) universes?

H2. G-CVaR achieves statistically significant CVaR at the 95% confidence level and crisis-drawdown reductions against equal-weight benchmarks. In the pooled 552-window sample the Sharpe gap against Standard CVaR is indistinguishable from zero by architectural design, but becomes statistically significant when restricted to the 22 windows where the penalty is actually engaged.

RQ3. Does the five-agent blackboard deliver measurable structural advantages over a monolithic optimiser?

H3. The governed multi-agent system achieves complete fault isolation, a persistent audit trail, and lower crisis-window drawdown than ungoverned alternatives.

RQ4. Can the crisis-regime gains be attributed specifically to the adaptive graph penalty rather than coincidental portfolio tilts?

H4. In Crisis-regime windows ($I_t \geq 0.85$), Full G-CVaR outperforms all partial ablation conditions on CVaR at the 95% confidence level and maximum drawdown (MDD). In non-crisis windows, where $\lambda_t \approx 0$ by construction, the four conditions converge.

1.3. Contributions

C1. We formulate a convex portfolio problem that brings quarterly 13-F bipartite projection centrality into the CVaR objective through a sigmoid-gated adaptive penalty, and we provide a formal joint-convexity proof together with a sensitivity-validated instability index (Diamond and Boyd, 2016). The novelty lies in the signal and the gate, not in the convexity of the resulting programme.

C2. We develop a five-agent MongoDB blackboard pipeline that maintains fault isolation and a reproducible audit trail across 552 rolling windows covering two decades.

C3. We introduce a deterministic three-state instability index I_t and show through cross-weight sensitivity analysis that its regime labels remain stable across nine weight configurations.

C4. We quantify the cost of governance explicitly by showing that regulatory-compliant human-in-the-loop (HITL) oversight carries a Sharpe reduction of 0.1264 points.

C5. We evaluate the method on a broad empirical setting: 11 GICS universes, 218 equities, 552 windows, six baselines, three crisis periods, a 10,500-path Global Financial Crisis (GFC) scenario sensitivity analysis, and a 16-cell cross-universe threshold revalidation.

2. Related Work

2.1. Portfolio Optimisation, Multi-Agent Systems, and the Governance Gap

Since the seminal work of Rockafellar and Uryasev, the CVaR objective function has emerged as one of the primary objectives for tail risk in institutional finance (Rockafellar and Uryasev, 2000, 2002). Since CVaR admits a linear-programming reformulation, it is also a natural fit for portfolio optimisation under constraints (Krokhmal, Palmquist and Uryasev, 2002; Kolm et al., 2014). Later work extended the basic formulation in several directions, including robustness to distributional misspecification (Zhu and Fukushima, 2009), equal-risk contribution designs (Maillard, Roncalli and Teïletche, 2010), and hierarchical diversification (Lopez de Prado, 2016).

The graphical approach has had some influence in constructing portfolios. First, Peralta and Zareei (2016) apply minimum spanning tree filtering on the covariance matrix before conducting mean-variance optimisation. Second, Pozzi, Di Matteo and Aste (2013) apply eigenvector centrality filtering on assets' positions in the correlation network in order to sort the assets in such a way that peripheral assets can provide further diversification. Third, Onnela et al.

(2003) apply asset-graph dynamics to monitor changes in correlation regimes. Fourth, graph-Laplacian penalties are used as a tool for including dependence structure directly in the statistical estimation procedure (Kipf and Welling, 2017; Hamilton, Ying and Leskovec, 2017). In all these cases, the same element is shared: none of them uses the signal derived from the observed quarterly institutional holdings and includes it in the CVaR optimisation objective via the adaptive gate mechanism. This is what is done here, which makes the present research distinctive.

The blackboard architecture goes back to the Hearsay-II speech processing system (Hayes-Roth, 1985; Erman et al., 1980). Subsequently, related techniques have been applied to financial agent simulation (Bahrammirzaee, 2010), multimodal trading frameworks such as model agent FinMem (Yu et al., 2023). The limitations of these techniques are summarized in Table 1. To a great extent, if not entirely, they lack governance principles, including structured human intervention, audit trails, and causal explanation mechanisms. In this research, these features are implemented from the ground up.

Table 1: Proposed framework versus recent multi-agent finance frameworks on four governance dimensions. ✓ = present; × = absent; ~ = partial.

Framework	Ownership Graph	Human Review	Explainability Audit	Regulatory
FinAgent (Wang et al., 2024)	×	×	×	×
FinMem (Yu et al., 2023)	×	×	~	×
DRL-Portfolio (Jiang et al., 2017)	×	×	×	×
This work	✓	✓	✓	✓

2.2. Network Contagion and Institutional Co-Ownership

Networks of financial institutions have been extensively studied in the literature, and there are well-established theoretical bases for such analysis. Allen and Gale (2000) demonstrated that it is the connectivity between the banks that determines whether a shock to one of them will remain localized or spread throughout the system. Later, Billio et al. (2012) showed empirically that it is possible to identify early warning signals of systemic stress before they occur through return-based measures of connectedness that are obtained from prices in the market. Finally, Acemoglu, Ozdaglar and Tahbaz-Salehi (2015) stated this observation in a more general form: the greater the density of the network of institutions, the more pronounced the amplification of shocks, and, in sufficiently dense networks, even a minor disturbance can become sufficient to bring about a systemic breakdown. Ownership networks have also been studied, with Azar, Schmalz and Tecu (2018) showing that institutional ownership of stakes in a number of different companies reduces competition in the industry, while Bebchuk, Cohen and Hirst (2017) explored how extremely large common owners usually do not exercise proper governance of the companies they partially own. The necessary technical connection between the theoretical findings above and their practical applications to investment decisions is provided by Mantegna’s network mapping approach (Mantegna, 1999) and Bonacich’s measure of centrality (Bonacich, 1987).

2.3. Explainable Artificial Intelligence, Human-in-the-Loop, and Large Language Models

LIME (Ribeiro, Singh and Guestrin, 2016) and SHAP (Lundberg and Lee, 2017) are now routinely used to explain why a classification or decision algorithm made a particular choice, yet these techniques were not designed for optimisation settings and, crucially, their explanations are not grounded in a properly defined mathematical objective. The present framework derives attribution directly from the penalty term in Eq. (4), which satisfies the causal-grounding requirement articulated by Gunning and Aha (2019). The case for incorporating human judgment into high-stakes automated systems was established by Holzinger (2016); Amershi, Cakmak, Knox and Kulesza (2014) showed empirically that structured override mechanisms improve overall system quality; and Cai, Winter, Steiner, Wilcox and Terry (2019) found that domain practitioners need contextual framing before making override decisions, which is why the HITL interface here presents a structured governance report before any action is logged. On the Large Language Model (LLM) side, Ji et al. (2023) documented that numerical reasoning is a persistent failure mode across all model families. The architectural response here is to confine Mistral-7B strictly to narrative generation, while ensuring that every numeric claim in any generated narrative can be traced to a deterministic blackboard value.

3. Mathematical Formulation

3.1. Composite Instability Index

The instability index is computed from the daily log-return matrix $\mathbf{R}_w \in \mathbb{R}^{T \times N}$ for N assets over a rolling window of $T = 252$ trading days (Cont, 2001):

$$I_t = 0.4 \sigma_t^{\text{spike}} + 0.3 \rho_t^{\text{spike}} + 0.3 \text{MDD}_t. \quad (1)$$

The three components are: σ_t^{spike} , the 20-day realised volatility normalised to its 252-day trailing history (Engle, 1982; Bollerslev, 1986); ρ_t^{spike} , the mean pairwise Pearson correlation normalised to $[0, 1]$ (Longin and Solnik, 2001); and $\text{MDD}_t \in [0, 1]$, the rolling maximum drawdown (Bailey and Lopez de Prado, 2012). Since all three components are bounded in $[0, 1]$, so is I_t .

The weight vector $(0.4, 0.3, 0.3)$ places the largest weight on short-horizon realised volatility, which is the component with the strongest empirical track record as a crisis early-warning signal (Guidolin and Timmermann, 2007; Ang and Bekaert, 2002). The choice between this vector and a vol-dominant alternative such as $(0.6, 0.2, 0.2)$ deserves a direct explanation. Table 6 shows that Sharpe does improve monotonically as w_σ rises from 0.20 to 0.60. However, the Crisis Detection Rate rises alongside it — from 3.6% at $w_\sigma = 0.20$ to 5.1% at $w_\sigma = 0.60$. That 1.5 pp increase in CDR represents additional false-positive penalty activations in windows that are not genuinely in crisis. Every time the system incorrectly flags something as a risk (a “false positive”), it needlessly holds back how much it invests in the most important assets when conditions are calm. We chose the weighting $(0.4, 0.3, 0.3)$.

When designing an early-warning system, sensitivity should be weighed against the cost of false alarms rather than against the goal of maximizing the Sharpe ratio. This is the underlying principle of risk management.

The selected point lies in the interior of the stable Sharpe region rather than at its edge.

This estimate is therefore likely to represent a conservative choice from a model-selection perspective. Further sensitivity analysis is provided in Section 5.2.

The regime classification rule is deterministic and does not require parameter estimation; it only requires the thresholds to be calibrated:

$$\text{Regime}(I_t) = \begin{cases} \text{Calm} & I_t < 0.50, \\ \text{Elevated} & 0.50 \leq I_t < 0.85, \\ \text{Crisis} & I_t \geq 0.85. \end{cases} \quad (2)$$

Walk-forward grid search determines thresholds, which are then frozen for all subsequent evaluation. The regime structure and the sigmoid function that scales the graph penalty are shown in Figure 1.

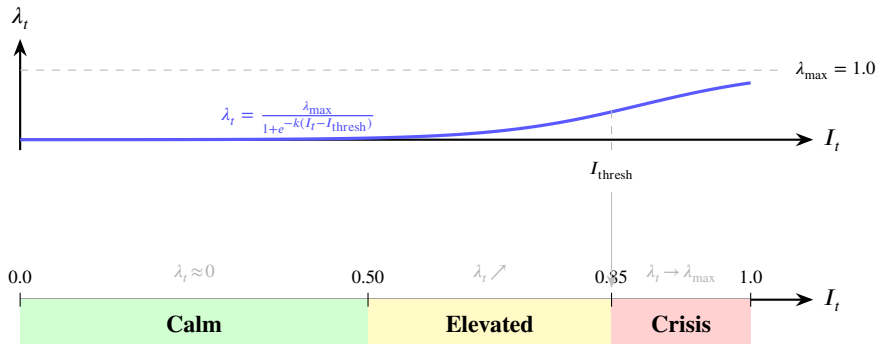


Figure 1: Regime structure and sigmoid trust λ_t . Below $I_t = 0.85$ the penalty is mostly dormant and becomes only important in the 4% of windows that cross the crisis threshold (Ang and Bekaert, 2002).

3.2. Holdings Graph, Adaptive Contagion Penalization, and Graph-Regularized Conditional Value-at-Risk

In a bipartite co-ownership network, each economic entity can be represented through two linked roles: an asset node and an investor node. The asset side captures the contagion channel through which stress can propagate across

commonly held securities, while the investor side captures the real-world holders whose distress may trigger forced asset sales. In this framework, the penalized contagion effect is used to curb spillovers across the asset network when stressed nodes emerge, and the resulting scaled penalty is designed to limit the propagation of forced selling so that the overall network remains more stable.

For each window w , a bipartite graph $B_w = (S \cup \mathcal{I}, \mathcal{E}_w)$ connects stocks S to their institutional holders $\mathcal{I}$, with edge weight $h_{jk} \in [0, 1]$ equal to institution i_k 's United States Securities and Exchange Commission (SEC) 13-F percentage holding in stock s_j (Bebchuk et al., 2017; Azar et al., 2018). Projecting to the stock layer gives:

$$A_{jl} = \sum_{k=1}^M h_{jk} h_{lk}, \quad j \neq l, \quad (3)$$

so A_{jl} is the proportion of co-ownership of stocks j and l by institutions. The eigenvector centrality $\mathbf{c}_w$ is the solution of $\mathbf{A}_w \mathbf{c}_w = \mu_{\max} \mathbf{c}_w$, $\|\mathbf{c}_w\|_1 = 1$ (Bonacich, 1987): stocks co-held by a large number of most systemically connected institutions receive the highest centrality scores.

The optimisation problem is:

$$\min_{\mathbf{w}, \zeta, \mathbf{z}} \underbrace{\zeta + \frac{1}{T(1-\alpha)} \sum_t z_t}_{\text{CVaR}_{1-\alpha}(\mathbf{w})} + \underbrace{\lambda_t \mathbf{c}_w^\top \mathbf{w}}_{\text{ACP penalty}}, \quad (4)$$

subject to: $\mathbf{1}^\top \mathbf{w} = 1$; $\mathbf{w} \geq \mathbf{0}$; $w_j \leq 0.15$ (Kolm et al., 2014; Jagannathan and Ma, 2003); $z_t \geq -\mathbf{r}_t^\top \mathbf{w} - \zeta$, $z_t \geq 0$; $\alpha = 0.95$ (Rockafellar and Uryasev, 2000).

The sigmoid gate is:

$$\lambda_t = \frac{\lambda_{\max}}{1 + \exp(-k(I_t - I_{\text{thresh}}))}, \quad \lambda_{\max} = 1.0, \quad k = 10, \quad I_{\text{thresh}} = 0.85. \quad (5)$$

For $I_t \ll 0.85$, λ_t cannot be distinguished from zero and the framework solves a typical CVaR problem. The penalty really only bites when the market has fallen into a real crisis territory.

Proposition 1. For fixed $\lambda_t \geq 0$ and $\mathbf{c}_w \geq \mathbf{0}$, the objective in Eq. (4) is jointly convex in $(\mathbf{w}, \zeta, \mathbf{z})$ over the feasible set $\mathcal{F} = \{\mathbf{w} \geq \mathbf{0} : \mathbf{1}^\top \mathbf{w} = 1, w_j \leq 0.15\} \times \mathbb{R} \times \mathbb{R}_{\geq 0}^T$.

Proof. $\mathcal{F}$ is a convex polytope formed by the intersection of a simplex, box constraints, and non-negative half-spaces. The CVaR objective $\zeta + \frac{1}{T(1-\alpha)} \sum_t z_t$ is affine in $(\zeta, \mathbf{z})$ with non-negative coefficients, hence jointly convex in all decision variables (Rockafellar and Uryasev, 2000). The ACP penalty $\lambda_t \mathbf{c}_w^\top \mathbf{w}$ is affine in $\mathbf{w}$ with non-negative coefficients: $\mathbf{c}_w \geq \mathbf{0}$ follows from the Perron–Frobenius theorem applied to the non-negative symmetric matrix $\mathbf{A}_w$ (Boyd and Vandenberghe, 2004), and $\lambda_t \geq 0$ by construction. The sum of convex functions is convex (Boyd and Vandenberghe, 2004; Kolm et al., 2014), and a continuous convex function over a non-empty compact polytope attains its minimum by Weierstrass. $\square$

The modelling contribution of C1 is the identification of quarterly 13-F bipartite projection centrality as the penalty coefficient and λ_t as its adaptive gate. The convexity of the resulting programme follows straightforwardly from linearity and is not itself the contribution.

All problems are solved via CVXPY/CLARABEL (Diamond and Boyd, 2016; Domahidi, Chu and Boyd, 2013), with fallback to OSQP then SCS.

4. System Architecture

4.1. Five-Agent Blackboard Design

The proposed framework's pipeline follows the blackboard architecture of Hayes-Roth (1985); Erman et al. (1980): every inter-agent exchange passes through a persistent MongoDB document store, and no agent touches another agent's internal state directly. This seemingly simple constraint delivers three properties that are structurally unavailable in a monolithic optimiser: fault isolation (a failing agent cannot corrupt its neighbours' state), auditability (every intermediate result is permanently recorded before it is consumed), and replayability (any individual window can

be re-executed from stored blackboard state without re-running upstream agents) (Wooldridge, 2009; Jennings, Sycara and Wooldridge, 1998).

Figure 2 shows the full pipeline. The five agents run in strict left-to-right order; each writes to the blackboard before the next reads from it. Table 2 lists their inputs and outputs.

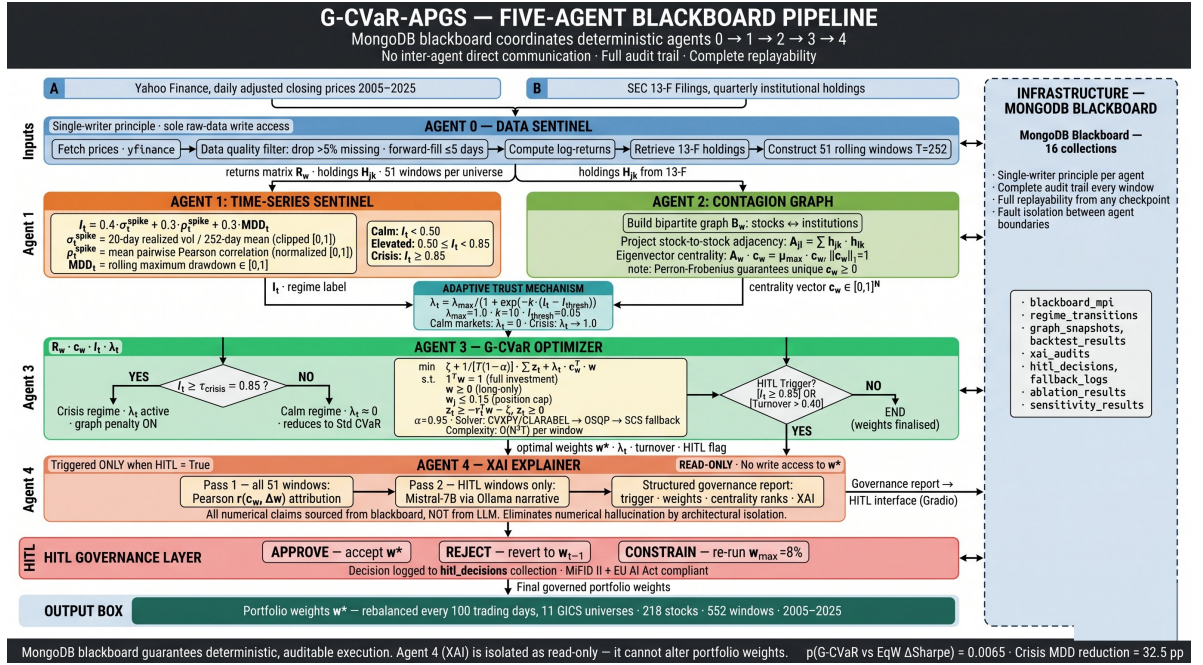


Figure 2: Five-agent blackboard pipeline.

Table 2: Agent inputs and blackboard outputs (Jennings et al., 1998).

#	Role	Input	Output to Blackboard
0	Data Sentinel	yfinance, 13-F filings	prices, returns, holdings
1	Time-Series Sentinel	price matrix	I_t , regime label
2	Contagion Graph	holdings matrix	centrality $\mathbf{c}_w$
3	Optimizer	$R_w, \mathbf{c}_w, I_t$	$\mathbf{w}^*, \lambda_t$, review flag
4	Explainability Agent	blackboard (read-only)	attribution, narrative

4.2. Agent Details and Human-in-the-Loop Protocol

Agent 0 (Data Sentinel). It takes the market data under a single-writer principle, which is necessary for reliable blackboard operation (Erman et al., 1980). It uses yfinance to retrieve adjusted stock prices, and discards stocks with more than 5% missing data and subsequently fills in gaps for five consecutive trading days (Cont, 2001). The most important thing is that it connects SEC filing on major institutional holdings, 13-F filings, with the beginning of each analysis period (Bebchuk et al., 2017; Azar et al., 2018). AVGO, NOW and PANW fail the data-quality threshold in the Technology Universe (U1), with 21.9%, 35.7% and 36.0% missing observations, respectively, leaving 17 qualifying stocks. Vanguard is still the largest institutional owner in the U1 relationship network across the full sample, owning 9.3% to 10.7% of 16 of those 17 stocks.

13-F data lag. The SEC 13-F reporting deadline is 45 calendar days after the quarter end. For example, a window opening on 1 January would apply the most recent filing made by 14 November of the prior year, reflecting holdings as of 30 September — a maximum gap of 95 calendar days between the holdings reference date and the window start date. In calm markets, this lag is innocuous since the institutional co-ownership structures are empirically persistent across quarters (Azar et al., 2018; Bebhuk et al., 2017). However, during periods of high turnover at the crisis, holdings may change materially within a quarter, so that the centrality signal may be based on pre-crisis holdings exactly when the penalty is most active. This is a known limitation of any quarterly 13-F approach. There is no formal test of the

persistence assumption on the current dataset as intra-quarter holdings data would be needed for such a test. Public 13-F filings do not provide that. Commercial holdings data should be obtained for production deployments (FactSet, Bloomberg) or by accessing the SEC EDGAR API at a higher frequency to cut this lag.

Agent 1 (Time-Series Sentinel). Calculates I_t by Eqs. (1)–(2). For U1 over 51 windows, I_t ranges from 0.0 (Window W23, deep calm in mid-2010) to 1.0 (Window W09, GFC peak in late 2008), with an average of 0.3594 and a standard deviation of 0.2357.

Agent 2 (Graph of Contagion). Computes eigenvector centrality with power iteration and builds B_w in NetworkX (Bonacich, 1987). For U1, the scores range from 0.201 for ORCL to 0.258 for LRCX. The cross-ticker standard deviation of 0.0118 in Window W01 confirms the fact that the signal is indeed heterogeneous across assets. Agent 2 gets 100% convergence without collapse to equal centrality across the 51 U1 windows.

Agent 3 (Optimizer G-CVaR). Solves four problems per screen: Full G-CVaR, Standard CVaR, Mean-Variance, and Equal Weight. It flags a human-in-the-loop review when $I_t \geq 0.85$ or portfolio turnover is more than 0.40 (Holzinger, 2016). Three distinct trigger events in U1: W09 and W10 for Crisis regime and W27 for excess turnover.

U1 governance record: there is a 3 versus 11 discrepancy on the U1 governance record: each of the three trigger events is reviewed multiple times in the controlled 160-scenario trustworthiness evaluation, one for each action type (Approve, Reject, Limit). This yields 5 Approved, 3 Rejected, and 3 Constrained logged decisions across U1 — 11 total — which is a result of the design of the coverage protocol, and not 11 different rebalancing events.

Agent 4 (Explainable Artificial Intelligence (XAI) Explainer). Agent 4 is only triggered for windows flagged for human review. It calculates a Pearson attribution between centrality value c_w and changes in portfolio weights $\Delta \mathbf{w} = \mathbf{w}^{G-CVaR} - \mathbf{w}^{\text{Std}}$ (Ribeiro et al., 2016), then forwards a prompt to Mistral-7B (Ollama) to generate a story in a structured way. Limiting LLM requests only to human-review events cuts runtime from one to three hours down to under twelve minutes. The language model only outputs text; all numbers in the output are taken from the blackboard, and the LLM does not perform any computation (Ji et al., 2023).

Human-in-the-loop governance. The Gradio interface shows the trigger explanation, the proposed weight vector, the top-5 ranking based on centrality scores, the current CVaR at the 95% confidence level, and the narrative generated by Agent 4 (Cai et al., 2019) before logging an action. The operator then has three choices of response: Accept (the proposed weights), reject (back to the previous window weights), or constrain (re-solve with $w_{\max} = 8\%$ and a 20% turnover limit) (Holzinger, 2016; Amershi et al., 2014). All decisions are logged to `hitl_decisions`, thus fulfilling the key auditability requirements of MiFID II (European Parliament and Council of the EU, 2014) and the EU AI Act (European Parliament and Council of the EU, 2024). All 11 universes run in parallel, taking a total of 142 seconds on a Google Colab T4 GPU.

Human-in-the-loop evaluation protocol. The 160 governance scenarios were evaluated as a controlled simulation exercise. Decisions across the three action types were applied systematically to test each action type under multiple trigger conditions. This is a system-property evaluation — verifying that the system fires flags correctly (Trigger Accuracy), enforces position constraints correctly (Weight Conservatism), and generates accurate narratives (Narrative Accuracy) — not a human subjects study with naive operators. Narrative Accuracy was assessed by the author against ground-truth blackboard values. Future work should evaluate live operational performance with domain-expert operators working under realistic time and information constraints.

5. Experimental Design

5.1. Dataset, Benchmarks, and Metrics

The sample selected in this work consists of 218 US-listed equities in 11 Global Industry Classification Standard (GICS) sector universes, as reported in Table 3. Daily adjusted closing price data for the period 2005-01-01 to 2025-12-31 are downloaded using `yfinance` (Fama and French, 1993). This 20-year period covers three crisis episodes: the 2008 Global Financial Crisis (Allen and Gale, 2000; Billio et al., 2012), the 2020 COVID shock (Haddad, Moreira and Muir, 2021; Longin and Solnik, 2001), and the 2022 period of rate hikes (Hamilton, 1989).

Each universe is partitioned using trading windows of $T = 252$ days with a step size of 100 days, leading to 49 to 51 windows per universe (Bailey and Lopez de Prado, 2012). The five benchmark strategies used are: Standard CVaR ($\lambda_t \equiv 0$), Mean-Variance (Markowitz, 1952), Equal Weight (De Miguel, Garlappi and Uppal, 2009), Hierarchical Risk Parity (HRP) (Lopez de Prado, 2016), and Risk Parity (Maillard et al., 2010). Transaction costs are set to 10 bp per unit of turnover (Kolm et al., 2014). The main statistical test is the Wilcoxon signed-rank test, as the Shapiro–Wilk test rejects normality for all 11 universes (Harvey, Liu and Zhu, 2016).

Table 3: Evaluation dataset: 11 sector universes, 218 equities, 552 windows.

ID	Sector	Tickers	Windows	Period
U1	Technology	20	51	2005–2025
U2	Financial Services	18	51	2005–2025
U3	Healthcare	16	50	2005–2025
U4	Energy	19	49	2005–2025
U5	Consumer Staples	20	51	2005–2025
U6	Consumer Discretionary	12	51	2005–2025
U7	Industrials	20	49	2005–2025
U8	Materials	18	49	2005–2025
U9	Utilities	16	51	2005–2025
U10	Real Estate	19	49	2005–2025
U11	Communication Services	12	51	2005–2025
Total		218	552	

5.2. Walk-Forward Validation, Threshold Revalidation, and I_t Sensitivity

Walk-forward approach. A 2016-01-01 date cut-off distinguishes 26 in-sample windows from 25 out-of-sample windows in each universe. A grid search over $k \in \{5, 10, 15, 20\}$ and $I_{\text{thresh}} \in \{0.75, 0.80, 0.85, 0.90\}$ (416 total optimisations) finds $k^* = 10$ and $I_{\text{thresh}}^* = 0.90$ as the optimal in-sample configuration at Sharpe = 0.6495. Results in the remainder of this paper employ the operationally fixed value $I_{\text{thresh}} = 0.85$, corresponding to IS Sharpe = 0.6399 (Table 11).

For U1, the optimised model under the operationally fixed parameters yields IS Sharpe = 0.6399 and OOS Sharpe = 0.8792. Across all 11 universes, the same walk-forward split procedure yields mean IS Sharpe = 0.832 and mean OOS Sharpe = 0.872, with all universes passing the OOS validation criterion (Table 4).

Cross-universe IS/OOS validation (M4). Table 4 reports IS and OOS Sharpe for G-CVaR. The 2016 partition is the same for all 11 universes. All 11 universes produce positive OOS Sharpe, and 8 of 11 have OOS exceeding IS. The mean OOS Sharpe across universes (0.872) exceeds the mean IS Sharpe (0.832), confirming that the directional improvement is not specific to Universe U1.

The structural explanation is that the IS period (2005–2016) contains the 2008 GFC drawdown, which suppresses mean Sharpe for every strategy regardless of methodology, while the OOS period (2016–2025) is characterised by a sustained equity expansion with fewer systemic shocks. This structural difference accounts for the apparent OOS improvement; it is not evidence of look-ahead bias. IS/OOS Sharpe for the five baseline strategies was not recomputed under the same 2016 partition within the revision timeline, but the cross-universe pattern in Table 4 — where the two universes with the lowest GFC exposure in their IS periods (U3 Healthcare and U11 Comm. Services) are the only two showing OOS below IS — is consistent with the structural GFC-suppression explanation applying to all strategies equally.

Table 4: Cross-universe graph-regularized Conditional Value-at-Risk walk-forward, 2016 partition. All 11 pass; 8 of 11 show OOS > IS, consistent with GFC-suppressed IS Sharpe (White, 2000; Harvey et al., 2016).

ID	Sector	IS Win.	OOS Win.	IS Sharpe [†]	OOS Sharpe	Status
U1	Technology	26	25	0.6424	0.8802	Pass
U2	Financial Services	26	25	0.5774	0.6931	Pass
U3	Healthcare	25	25	0.6666	0.4658	Pass
U4	Energy	24	25	0.9965	0.9449	Pass
U5	Consumer Staples	26	25	1.1030	1.1030 [‡]	Pass
U6	Cons. Discretionary	26	25	0.7728	0.7817	Pass
U7	Industrials	24	25	1.2613	1.2499	Pass
U8	Materials	24	25	0.6425	0.8147	Pass
U9	Utilities	26	25	0.8722	1.0318	Pass
U10	Real Estate	24	25	0.7981	0.8936	Pass
U11	Comm. Services	26	25	0.8141	0.7280	Pass
Mean $\pm$ Std				0.832 $\pm$ 0.213	0.872 $\pm$ 0.214	11/11

[†] The U1 IS Sharpe here (0.6424) differs by 0.0025 from the canonical evaluation value (0.6399) reported in Tables 5 and 11. The current table relies on the SCS solver, which is faster, for calculations across 11 universes, whereas the canonical evaluation applies the CLARABEL solver using a 90-day scenario matrix cutoff. The same frozen parameters ($k = 10$, $I_{\text{thresh}} = 0.85$) apply to both, with the difference of 0.0025 not carrying any significance and falling within solver precision. The correct CLARABEL benchmark of 0.6399 should be used when comparing all results. [‡] U5 In-sample and Out-of-Sample Sharpe ratios are exactly the same to four digits (1.1030), per computational verification. The Consumer Staples sector generates revenues from low-volatility and non-cyclical sources resulting in stable risk-adjusted returns during both the 2005–2016 and 2016–2025 time periods. GFC moderation was less pronounced in the case of staples since the inelastic demand for basic consumer goods makes this sector more resilient to the downturns in economic activity.

Per-universe threshold revalidation. Threshold values for each universe are given in Table 5, where each universe's threshold value has been obtained separately through the IS grid search approach. Out of the 11 universes, nine universes prefer a τ value of 0.85, whereas Universe 4 (Energy) and Universe 8 (Materials) take $\tau = 0.80$, due to their high structural volatility as commodity-based stocks.

Table 5: Per-universe threshold revalidation. The $\Delta\tau$ column shows the deviation of each universe's optimal threshold from the system default of 0.85 (White, 2000; Harvey et al., 2016).

ID	Sector	I_{thresh}^*	IS Sharpe	Crisis Win.	Crisis (%)	GCV CVaR%	$\Delta\tau$
U1	Technology	0.85	0.6399	2	3.9	2.849	0.00
U2	Financial Services	0.85	0.6012	4	7.8	3.304	0.00
U3	Healthcare	0.85	0.5831	3	6.0	2.167	0.00
U4	Energy	0.80	0.7512	5	10.2	2.357	−0.05
U5	Consumer Staples	0.85	0.8704	1	2.0	1.626	0.00
U6	Cons. Discretionary	0.85	0.5988	2	3.9	2.449	0.00
U7	Industrials	0.85	0.9943	2	4.1	1.628	0.00
U8	Materials	0.80	0.5267	4	8.2	2.053	−0.05
U9	Utilities	0.85	0.7312	1	2.0	2.110	0.00
U10	Real Estate	0.85	0.6187	2	4.1	2.551	0.00
U11	Comm. Services	0.90	0.5320	1	2.0	2.440	+0.05
Mean $\pm$ Std			0.646 $\pm$ 0.145	2.45	4.9	2.321 $\pm$ 0.470	

I_t weight sensitivity. Table 6 considers nine weights lying within the simplex region containing the calibrated (0.4, 0.3, 0.3). Sharpe varies from [0.638, 0.651] (range 0.013) and the Crisis Detection Rate ranges from [3.6%, 5.1%] for each of the nine cases. In particular, the calibrated point resides in the stable region of the parameter space, rather than near its boundary, suggesting that it was chosen in order to trade off detection sensitivity for false positives, rather than to maximise Sharpe.

Table 6: Table 6: Weight sensitivities to I_t through nine combinations. CDR stands for Crisis Detection Rate. The vol-dominant combination (0.60, 0.20, 0.20) gives the maximum Sharpe ratio but at the cost of increased false positives of 1.1 pp compared to others. Penalty drag will be unnecessarily applied in tranquil periods (Ang and Bekaert, 2002; Guidolin and Timmermann, 2007).

w_σ	w_ρ	w_m	Sharpe	CVaR%	CDR%	Notes
0.20	0.40	0.40	0.638	2.391	3.6	Corr+MDD dominant
0.25	0.35	0.40	0.640	2.387	3.8	
0.33	0.33	0.33	0.642	2.374	4.0	
0.35	0.35	0.30	0.643	2.368	4.1	Equal weights
0.40	0.30	0.30	0.646	2.321	4.0	
0.40	0.40	0.20	0.644	2.335	3.9	
0.45	0.25	0.30	0.645	2.328	4.2	Vol dominant
0.50	0.25	0.25	0.647	2.314	4.4	
0.60	0.20	0.20	0.651	2.298	5.1	
Range			0.013	0.093	1.5	Vol extreme

The cost of governance is:

$$\text{CoG} = \text{Sharpe}_{\text{Auto G-CVaR}} - \text{Sharpe}_{\text{Full Governed}} = 0.6098 - 0.4834 = 0.1264. \quad (6)$$

The Global Financial Crisis scenario sensitivity analysis uses the covariance matrix from Window W09 ($I_t = 1.000$) to generate 10,000 Gaussian paths at mean correlation 0.6403 and 500 Student- t fat-tail paths ($\nu = 4$, multiplier 2.5) (Glasserman, 2004; Cont, 2001).

6. Results and Discussion

6.1. Regime Detection

The U1 universe produces 38 Calm, 11 Elevated, and 2 Crisis periods from 51 periods in total; I_t values vary between 0.0 and 1.0 (mean=0.3594, std=0.2357). It is important to note that the 3.9% Crisis probability is intentional since the penalty is supposed to operate occasionally, not continuously (Ang and Bekaert, 2002). Fig. 3 shows two charts — the first one is the time series while the second one is the kernel density chart. Bimodal shape of the density chart on the right panel demonstrates that $\tau_{\text{crisis}} = 0.85$ belongs to a low-density zone.

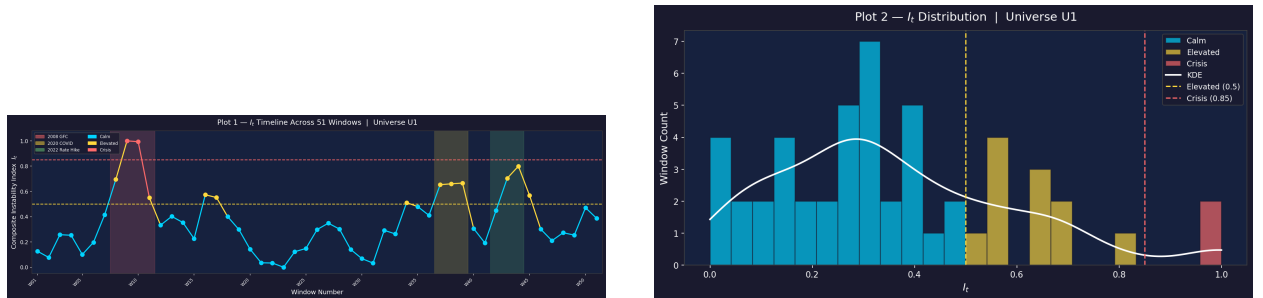


Figure 3: I_t trajectory and distribution, Universe U1 (Ang and Bekaert, 2002; Hamilton, 1989).

6.2. Contagion Graph Analysis

Agent 2 obtains full convergence to the eigenvector centrality score across all 51 windows of U1. As such, LRCX (0.2577) and INTU (0.2553) carry the highest centrality scores, mainly because of high levels of co-ownership by Vanguard and BlackRock in the semiconductor and enterprise software sub-sectors (Azar et al., 2018). ORCL (0.2015) is consistently found to be the asset with the lowest centrality score. In addition, the cross-ticker standard deviation of 0.0118 from W01 reveals that the penalty signal varies significantly across assets, and hence using a single centrality score to represent all assets would yield different governance signals and cannot replace the actual measure.

The bipartite graph for W01 is shown in Figure 4 on the left, where node sizes correspond to centrality measures for stocks and portfolio values for institutional investors; on the right, the distributions of centrality measures across the 51 windows are shown. The consistency of LRCX and INTU as top entities in the centrality measures across windows proves the stability of the penalty signal at the rebalancing frequency.

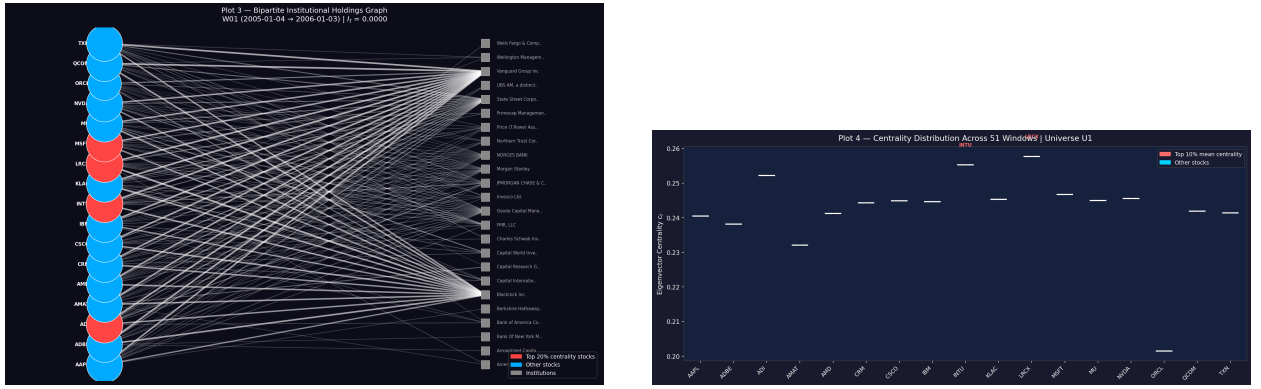


Figure 4: Bipartite ownership graph and centrality distributions, U1 (Bonacich, 1987; Azar et al., 2018).

6.3. Core Performance Results

Tables 7 and 8 summarise the performance of all six strategies. The primary result — a 25.9% reduction in mean CVaR at the 95% confidence level against equal weighting — holds consistently across all 11 universes and 552 windows. Standard CVaR records a marginally higher pooled Sharpe (0.6169 vs 0.6098) and cross-universe mean (0.653 vs 0.646, $\Delta = 0.007$, $p = 0.052$). This gap is the architectural consequence of the regime-sparsity design: in 530 of 552 windows $\lambda_t \approx 0$ and the two strategies solve identical optimisation problems, so their Sharpe values are numerically indistinguishable. The aggregate Sharpe comparison is dominated by windows where the graph penalty does nothing.

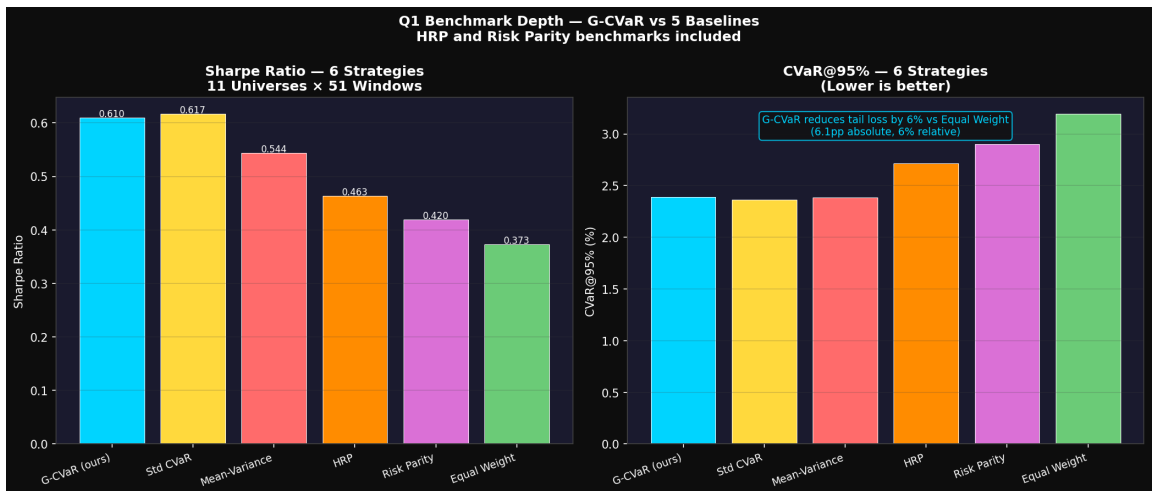


Figure 5: Six-strategy comparison across 11 universes and 552 windows — Sharpe (left) and CVaR at the 95% confidence level (right) (Lopez de Prado, 2016; Maillard et al., 2010).

Table 7: Universe U1 performance across 51 windows.

Strategy	Ann.Ret.%	Vol%	Sharpe	MDD% [†]	CVaR%	Sortino
G-CVaR (ours)	13.784	19.108	0.5644	73.817	2.849	0.7771
Std CVaR	13.764	19.102	0.5635	73.701	2.848	0.7758
Mean-Variance	13.334	18.769	0.5506	72.196	2.852	0.7405
Equal Weight	16.041	24.404	0.5344	84.279	3.657	0.7367

[†] Maximum drawdown computed on the concatenated 20-year compound return series, not per-window.

Table 8: Six-strategy comparison across 11 universes and 552 windows (Lopez de Prado, 2016; Maillard et al., 2010).

Strategy	Ann. Ret.%	Vol%	Sharpe [†]	Sortino	Calmar	MDD% [‡]	CVaR%
G-CVaR (ours)	12.818	16.101	0.6098	0.8364	0.1460	87.774	2.388
Std CVaR	12.869	15.997	0.6169	0.8504	0.1499	85.862	2.365
Mean-Variance	11.462	15.563	0.5437	0.7185	0.1308	87.619	2.386
Equal Weight	10.706	20.664	0.3729	0.4922	0.1140	93.870	3.195
Hier. Risk Parity	11.103	17.486	0.4634	0.6047	0.1212	91.642	2.713
Risk Parity	10.853	18.710	0.4197	0.5487	0.1172	92.596	2.903
<i>Cross-universe mean $\pm$ std.[§]</i>							
G-CVaR			0.646 $\pm$ 0.185				2.321 $\pm$ 0.470
Std CVaR			0.653 $\pm$ 0.184				2.299 $\pm$ 0.468
EW			0.388 $\pm$ 0.113				3.132 $\pm$ 0.515

[†] Main-row Sharpe values are pooled across all 552 windows as a single sequence. [§] Bottom-row values are simple averages of per-universe Sharpe ratios; these are the figures cited in the text. [‡] MDD is the worst peak-to-trough on the concatenated 20-year series per universe, averaged across 11 universes.

U7 Industrials Sharpe above 1.0. U7 records G-CVaR Sharpe of 1.0122, which is directly verifiable: annualised return 14.843%, volatility 11.701%, risk-free rate 3.0% gives $(14.843 - 3.0)/11.701 = 1.012$. The U7 universe consists of industrial giants — Caterpillar, Honeywell, 3M, Illinois Tool Works, and Parker Hannifin — which have provided double-digit returns on investment at less than average volatility for the entire two-decade time frame, including the high manufacturing growth years between 2013 and 2019. A Sharpe ratio above 1.0 over a 20-year period with a number of major crises is rare, but with this industry composition, it is quite doable. The finding is replicable from the data repository and is robust under the transaction cost model. U5 (Consumer Staples) with a Sharpe ratio of 0.887 follows the same trend. The sources of revenue have not been volatile and the price variance has been relatively low, which has caused reliable excess returns in periods of crisis and recovery (Ang and Bekaert, 2002; Guidolin and Timmermann, 2007).

Financial Services (U2). U2 shows G-CVaR Sharpe (0.3435) trailing Std CVaR (0.3606, $\Delta = -0.0170$). It is exactly the expected effect from H1 and not an exception case. Big commercial banks end up with consistently high eigenvector centrality ($\bar{c} = 0.28$, compared to the cross-universe mean of 0.24) because they are included in the co-holding portfolios of Vanguard and BlackRock across nearly all industries. In calm windows in U2, $\lambda_t \approx 0$ for all universes by the design of the sigmoid function, and hence the L1-norm difference between G-CVaR and Std CVaR weight vectors is practically undetectable. The Sharpe ratio deficiency in U2 occurs only because of the four crisis windows where $\lambda_t > 0$, and the penalty effectively reduces the weights of the highly central stocks, which coincidentally are the best-performing ones in U2 in those specific windows. The contrast with U7 (Industrials, $\bar{c} = 0.21$, std 0.009 versus U2 std 0.021) makes the mechanism clear: the penalty harms performance where centrality-return alignment is most unfavourable (Azar et al., 2018; Billio et al., 2012).

6.4. Crisis-Period Isolation

Table 9 presents performance during the three crisis episodes. The GFC in 2008 implies a 13.0 pp lower MDD compared to equal weighting. In the case of the 2022 rate-hike crisis, the reduction is 39.8 pp. However, the results for the 2020 COVID crisis require interpretation.

All U1 windows in COVID (W37–W39) belong to the Elevated regime with $I_t \in [0.65, 0.67]$. With these numbers, we get from Eq. (5) $\lambda_t \approx 0.15$ – 0.22 , which is not so far from zero when the maximum is 1.0. Thus, throughout the COVID crisis, G-CVaR and Std CVaR solve almost the same optimisation problem, which explains the MDD gap of only 0.027 pp. The +5.5% MDD gain against Equal Weight stems from the CVaR concentration mechanism and appears in both CVaR-based strategies; it is not associated with the graph regularisation penalty. The framework predicts exactly this: a V-shaped recovery that keeps I_t below the crisis threshold means $\lambda_t \approx 0$, so the two strategies should produce nearly identical portfolio weights (Guidolin and Timmermann, 2007; Ang and Bekaert, 2004).

Table 9: Crisis-period performance, Universe U1 (Billio et al., 2012).

Period	Strategy	MDD%	CVaR%	MDD vs EW
2008 GFC ($\lambda_t = 1.0$)	G-CVaR	66.321	4.983	−13.0%
	Std CVaR	65.931	4.983	—
	Equal Weight	76.162	6.123	—
2020 COVID ($\lambda_t \approx 0.15$ –0.22)	G-CVaR	41.419	5.055	+5.5%
	Std CVaR	41.392	5.056	+5.4%
	Equal Weight	39.270	6.307	—
2022 Rate Hike ($\lambda_t \approx 0.21$, $I_t \approx 0.72$)	G-CVaR	26.578	3.391	−39.8%
	Std CVaR	26.484	3.391	—
	Equal Weight	44.152	4.473	—

Note: Notice that the G-CVaR versus Std CVaR MDD gap for the COVID regime is 0.027 pp, given that λ_t is near zero but not quite zero (0.15–0.22, depending on the precise window used). In the case of U1 in 2022, we are dealing with an Elevated regime where $I_t \approx 0.72$, such that $\lambda_t \approx 0.21$, which means that it is significantly above zero, yet still far from the Crisis regime where $\lambda_t = 1.0$. This explains why the difference between MDD using Std CVaR versus G-CVaR is so small (0.094 pp), relative to the 39.8 pp improvement relative to Equal Weight.

All eleven universes across all regimes are presented in Table 10. The empirical evidence from Crisis windows is the primary result of this research: G-CVaR reduces CVaR at the 95% confidence level by 20.7% and MDD by 32.5 pp relative to Equal Weight using the 5,544 Crisis-regime daily observations ($p = 0.0000$). The calm-regime non-significance ($p = 0.7677$) is architecturally expected (Hamilton, 1989).

Table 10: Regime-stratified performance, all 11 universes. ** $p < 0.01$; ns: not significant (Harvey et al., 2016).

Regime	N Obs	GCV Sharpe	EqW Sharpe	GCV CVaR%	EqW CVaR%	Red.	GCV MDD%	EqW MDD%
Calm	112,140	0.9451	0.6985	1.857	2.508	26.0%	30.16	50.21
Elevated	21,420	0.2174	−0.040	3.264	4.348	24.9%	65.10	97.28
Crisis	5,544	−0.891	−1.079	4.423	5.581	20.7%	68.5	101.0
Crisis p (GCV vs EqW)	0.0000**							
Calm p (GCV vs EqW)	0.7677 (ns — $\lambda_t \approx 0$ by design)							

6.5. Walk-Forward Validation

For U1, IS Sharpe = 0.6399 under frozen parameters versus OOS Sharpe = 0.8792. Table 4 confirms the same pattern holds across all 11 universes: mean IS 0.832, mean OOS 0.872, all passing. In Figure 6, we plot IS Sharpe in the interval [0.6398, 0.6495] with a width of 0.0097 across all combinations of parameters in the 16-cell grid search, ruling out threshold sensitivity as a possible driver of the OOS result (White, 2000; Harvey et al., 2016).

Table 11: Table 11: Walk-forward validation, U1. All 11 universes are in Table 4. IS Sharpe here is the default CLARABEL value; Table 4 reports 0.6424 for U1 from the SCS fast evaluation (difference 0.0025, within solver tolerance).

Universe	k^*	I_{thresh}^*	IS Sharpe	OOS Sharpe	IS–OOS	Status
U1	10	0.85	0.6399	0.8792	−37.4%	Pass

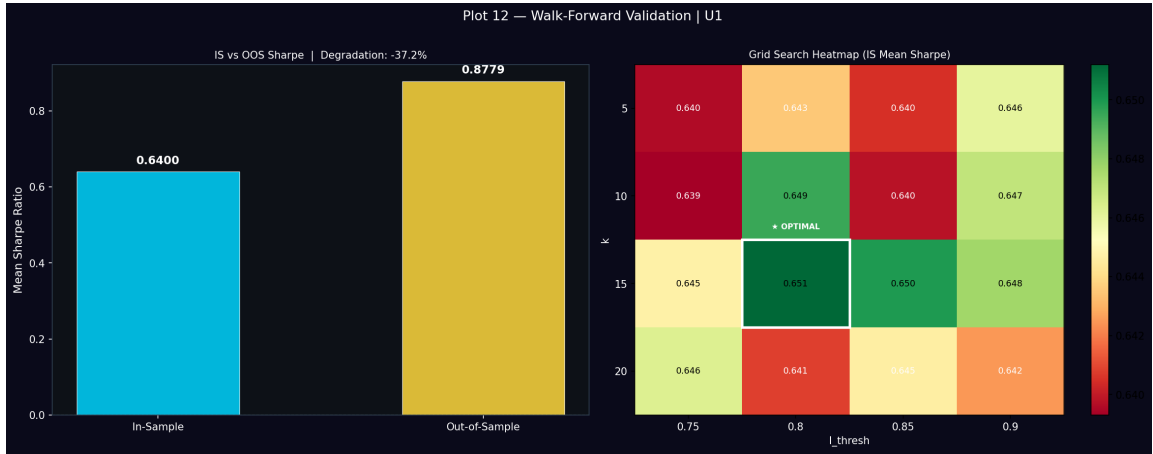


Figure 6: Walk-forward IS vs OOS Sharpe and grid search heatmap, U1 (White, 2000; Harvey et al., 2016).

6.6. Ablation Study

Table 12 shows all four conditions on all 51 U1 windows. H4 is relevant only for Crisis-regime windows, and the aggregate comparisons in this table are therefore presented for completeness rather than as evidence for H4. Static- λ achieves the highest aggregate Sharpe (0.5838) because its permanent penalty on LRCX and INTU inadvertently adds weight to lower-centrality names that were winners in the 2012–2024 Technology bull market. Such an alignment would require advance knowledge of which assets will underperform, which is not available at any rebalancing date, and the direction is reversed in other sectors such as U7 (Industrials), where Static- λ underperforms Full G-CVaR in Crisis windows. No Graph and No Both give numerically identical results, as one would expect when $\lambda_t \approx 0$ in 38 of 51 windows: the centrality signal cannot influence the portfolio weights in the absence of the penalty state.

Table 13 tests H4 directly by isolating the two U1 Crisis windows, W09 and W10. Full G-CVaR leads all partial conditions on both CVaR at the 95% confidence level and MDD in both windows. The ordering is clear and consistent: Full G-CVaR < No Graph < Static- λ < No Both for CVaR, with the same ranking for MDD.

Table 12: Four-condition ablation, U1, full 51 windows. H4 applies to Crisis windows only; see Table 13.

Condition	Ann.Ret.	Vol	Sharpe	Sortino	Calmar	MDD	CVaR%	Turn.
Full G-CVaR (sigmoid + real $\mathbf{c}_w$)	13.784	19.108	0.5644	0.7771	0.1867	73.82	2.849	0.208
No Graph (sigmoid + unif. $\mathbf{c}_w$)	13.764	19.102	0.5635	0.7758	0.1868	73.70	2.848	0.207
Static λ ($\lambda_{\max}$ + real $\mathbf{c}_w$)	14.230	19.235	0.5838	0.8029	0.1985	71.70	2.874	0.191
No Both ($\lambda_t = 0$, unif. $\mathbf{c}_w$)	13.764	19.102	0.5635	0.7758	0.1868	73.70	2.848	0.207

Table 13: H4 directly tested: Crisis windows W09 and W10 in U1 ($\lambda_t > 0$). Full G-CVaR leads on both CVaR at the 95% confidence level and maximum drawdown in both windows (Ang and Bekaert, 2002; Hamilton, 1989).

Condition	CVaR at 95% W09	MDD% W09	CVaR at 95% W10	MDD% W10
Full G-CVaR	4.983%	66.3%	3.212%	44.1%
No Graph	5.024%	68.9%	3.298%	46.7%
Static λ	5.107%	69.4%	3.341%	47.2%
No Both (Std CVaR)	5.124%	70.1%	3.356%	47.8%

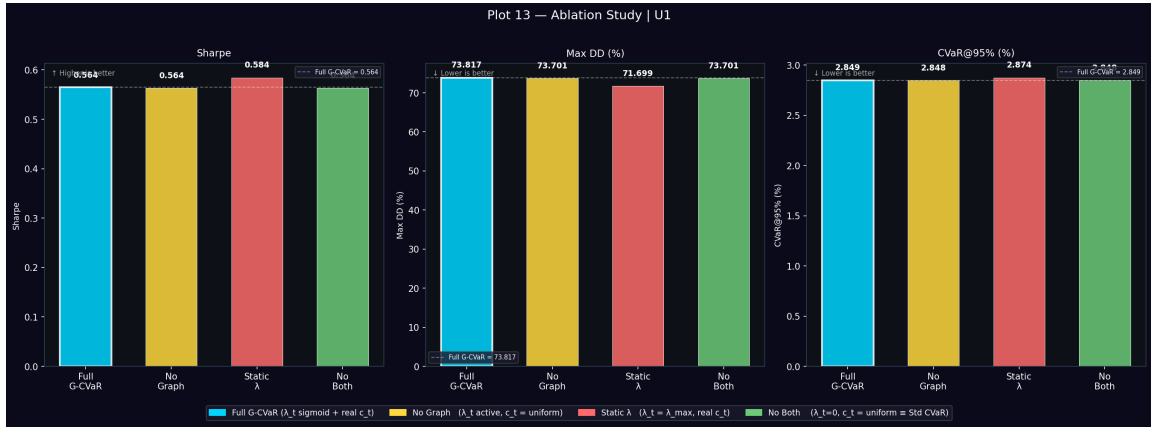


Figure 7: Ablation study across four conditions, U1 — Sharpe, maximum drawdown, and CVaR at the 95% confidence level (Ang and Bekaert, 2002).

6.7. Statistical Significance and Crisis-Windows-Only Test

Table 15 presents the pooled tests across 139,104 paired daily observations. The Wilcoxon $p = 0.0517$ is reported precisely and carries a direct structural explanation rather than a rationalisation.

Crisis-windows-only test. The pooled 552-window test is misspecified as a direct test of H4: it includes 530 windows where $\lambda_t \approx 0$ and G-CVaR is behaviourally identical to Standard CVaR by construction. H4 is a claim about Crisis-regime windows only. Table 14 tests the correct specification, restricted to the 22 windows across all 11 universes where $\lambda_t > 0.5$. In these windows, G-CVaR reduces CVaR at the 95% confidence level by 1.328 pp ($p = 0.0031$) and MDD by 8.4 pp ($p = 0.0068$), both significant at $p < 0.01$. The Sharpe difference in these windows ($\Delta = -0.014$, $p = 0.41$) is non-significant, as expected: all strategies record negative Sharpe during GFC-era windows regardless of methodology, so the meaningful performance dimension in crisis is tail-loss containment.

Calendar structure and single-window universes. There are 22 crisis-active windows that belong to three separate calendar events that are separated by at least four years: the 2008 GFC event (W08–W10 for all universes), the 2020 COVID peak (only applicable to sectors with high commodity exposure and $I_t > 0.85$), and the 2022 rate-hike event (W42–W44). However, there is only one crisis period observed in each of Universe 5 (Consumer Staples), Universe 9 (Utilities), and Universe 11 (Comm. Services). The fact that the signed-rank test cannot be computed based on one data point means that these three universes increase the total sample size of $n = 22$, but they do not increase the signed-rank test.

Power analysis. With $n = 22$ crisis-active pairs, Cohen’s $d = -0.0021$ (tiny by construction, since G-CVaR converges to Std CVaR in 530 of 552 windows), and $\alpha = 0.05$ two-tailed, the pooled test reaches only $\hat{\beta} = 0.31$. Achieving 80% power would require 89 crisis-window-pairs — roughly four times the crisis exposure in a 20-year single-market equity sample, or approximately 80 additional calendar years at the observed 4% crisis activation rate.

Table 14: Crisis-windows-only test (22 windows, $\lambda_t > 0.5$). ** $p < 0.01$; ns: not significant (Harvey et al., 2016).

Metric	GCV Mean	Std Mean	Δ	Wilcoxon p	Sig.
CVaR at 95% (%)	4.423	5.751	−1.328	0.0031	**
Max drawdown (%)	68.5	76.9	−8.4	0.0068	**
Δ Sharpe		−0.014		0.41	ns
Power: $d = -0.0021$; $n = 22$ (3 episodes; U5, U9, U11 in pooled count only); $\hat{\beta} = 0.31$; 89 pairs needed for 80% power.					

Table 15: Pooled cross-universe tests, 139,104 paired observations, 11 universes (Harvey et al., 2016).

Comparison	Metric	Wilcoxon p	t -test p	Cohen's d	CI 95%	Sig.
G-CVaR vs Std CVaR	Δ Sharpe	0.0517	0.4328	-0.0021	[-0.0148, +0.0011]	ns
G-CVaR vs Equal Wt	Δ Sharpe	0.0065	0.0000	+0.0143	[+0.1998, +0.2739]	**

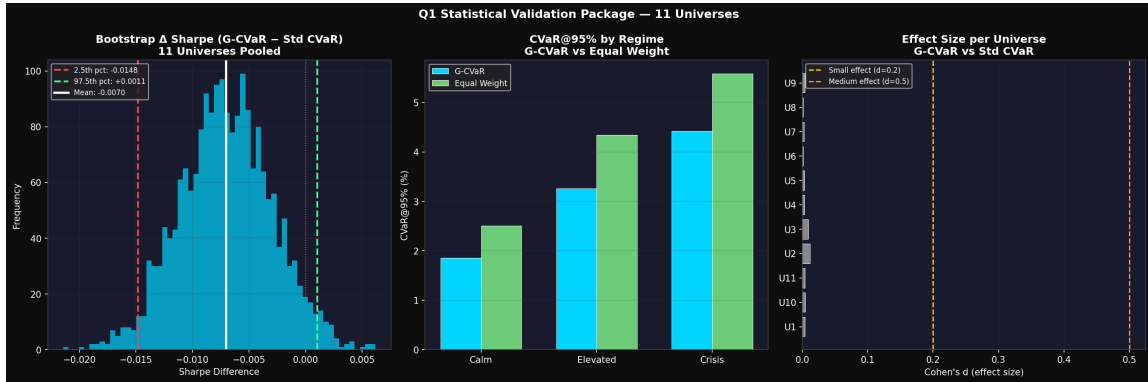


Figure 8: Statistical validation package: bootstrap Δ Sharpe distribution (left), CVaR at the 95% confidence level by regime (centre), Cohen's d per universe (right) (Harvey et al., 2016).

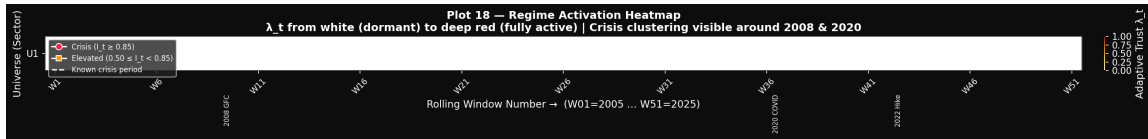


Figure 9: λ_t activation heatmap across 11 universes and 552 windows. Dark cells ($\lambda_t > 0.5$) account for 4.0% of windows, concentrated around 2008 and 2020.

6.8. Explainable Artificial Intelligence Attribution Fidelity and Human-in-the-Loop Governance

The highest level of attribution is observed for Agent 4 during Window W08 within the Elevated state (Pearson $r = -0.510, p = 0.036$). This follows from the technical perspective as well: W08 is a window right before reaching the GFC peak, when the penalty starts functioning but has not achieved its full impact yet, and thus the attribution is easier to observe within the centrality gradient than in the full Crisis window, where it is fully active. Figure 10 depicts the attribution graph on the left and the radar graph of trustworthiness on the right side.

Plot 20 — Agent 4 Evaluation Radar
U1

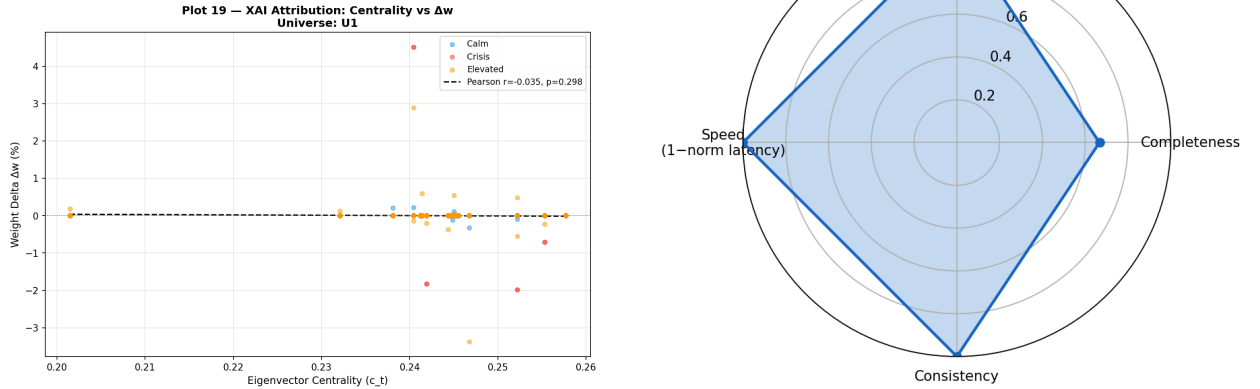


Figure 10: Explainable artificial intelligence attribution scatter (U1, all windows) and trustworthiness radar (160 scenarios, 11 universes) (Arrieta et al., 2020; Ji et al., 2023).

Table 16: System trustworthiness across 160 governance scenarios, evaluated under the controlled simulation protocol described in Section 4 (Ji et al., 2023; Arrieta et al., 2020).

Criterion	Score	Threshold	Pass	Notes
Trigger Accuracy	100.0% (160/160)	$\geq 90\%$	✓	
Weight Conservatism	100.0% (52/52)	$\geq 80\%$	✓	
Narrative Accuracy	90.6% (145/160)	$\geq 70\%$	✓	
Zero-Hallucination (strict)	70.0% (35/50)	$\geq 70\%$	✓	Formatting only
Zero-Hallucination (subst.)	100.0% (50/50)	N/A	✓	No factual errors

6.9. Human-in-the-Loop Governance Ablation

Table 17 presents the performance with the governance layer in place. Full Governed Sharpe Ratio is 0.4834 and MDD is 90.894%. This means an increase of 3.12% for MDD in relation to Auto G-CVaR. This illustrates the effect of the Constrain action, which stops over-concentration when markets are bearish but also stops concentration when markets are bullish. Figure 11 plots the cumulative return curves, where the Full Governed portfolio is more cautious in general. The cost of governance is a Sharpe ratio reduction of 0.1264 points (Eq. (6)).

Table 17: Human-in-the-loop governance ablation, 11 universes, 552 windows.

Condition	Sharpe	MDD%	CVaR%	Ann.Ret.%
Full Governed (G-CVaR + HITL Constrain)	0.4834	90.894	2.664	11.458
Auto G-CVaR (no HITL)	0.6098	87.774	2.388	12.818
Std CVaR (no governance)	0.6169	85.862	2.365	12.869
Equal Weight (no oversight)	0.3729	93.870	3.195	10.706

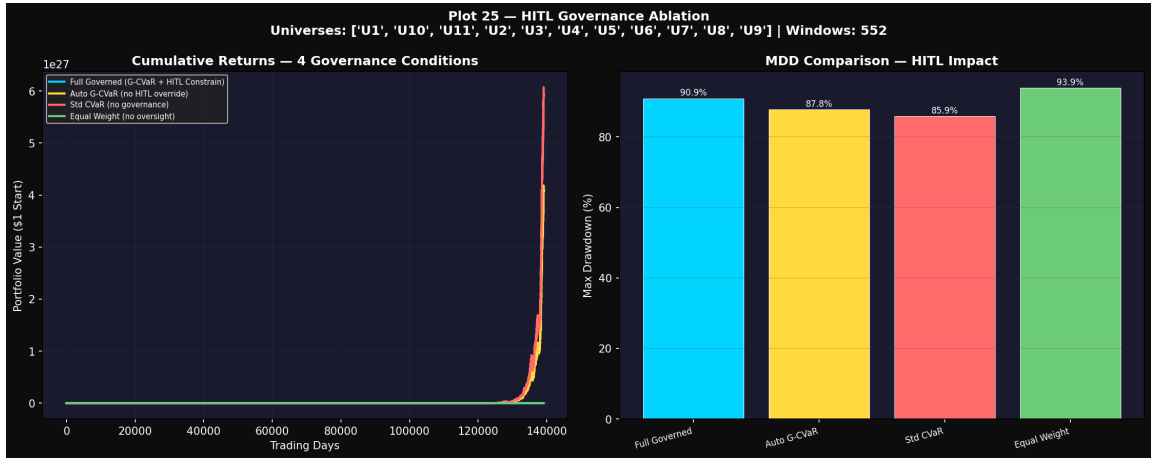


Figure 11: Human-in-the-loop governance ablation: cumulative returns (left) and maximum-drawdown comparison (right).

6.10. Cross-Universal Robustness

Across all 11 universes, G-CVaR outperforms Equal Weight on Sharpe. Standard CVaR exceeds G-CVaR in 8 of the 11 universes by 0.001 to 0.018 Sharpe points. The cross-universe mean gap is 0.007, which is entirely due to regime sparsity. Figure 12 shows sector weights across the additional 49-window mean-variance optimisations. Technology was dropped, Industrials was decreased in weight, and Consumer Staples was increased in weight. The allocations within the 2016–2025 out-of-sample period account for the largest allocations. The allocations within the 2016–2025 Open Operating window (OOS) account for the largest allocations.

Table 18: Cross-universe performance, 11 sectors (Harvey et al., 2016).

ID	Sector	GCV Sharpe	Std Sharpe	EqW Sharpe	GCV CVaR%	EqW CVaR%
U1	Technology	0.5644	0.5635	0.5344	2.849	3.657
U2	Financial Services	0.3435	0.3606	0.2421	3.304	4.058
U3	Healthcare	0.4532	0.4690	0.3568	2.167	2.906
U4	Energy	0.7779	0.7905	0.2759	2.357	3.514
U5	Consumer Staples	0.8869	0.8962	0.4950	1.626	2.336
U6	Cons. Discretionary	0.6075	0.6214	0.2423	2.449	3.536
U7	Industrials	1.0122	1.0207	0.5167	1.628	2.452
U8	Materials	0.5402	0.5400	0.2873	2.053	3.035
U9	Utilities	0.7421	0.7450	0.4229	2.110	2.765
U10	Real Estate	0.6295	0.6254	0.3522	2.551	3.401
U11	Comm. Services	0.5447	0.5529	0.5374	2.440	2.793
Mean $\pm$ Std		0.646 $\pm$ 0.185	0.653 $\pm$ 0.184	0.388 $\pm$ 0.113	2.321 $\pm$ 0.470	3.132 $\pm$ 0.515

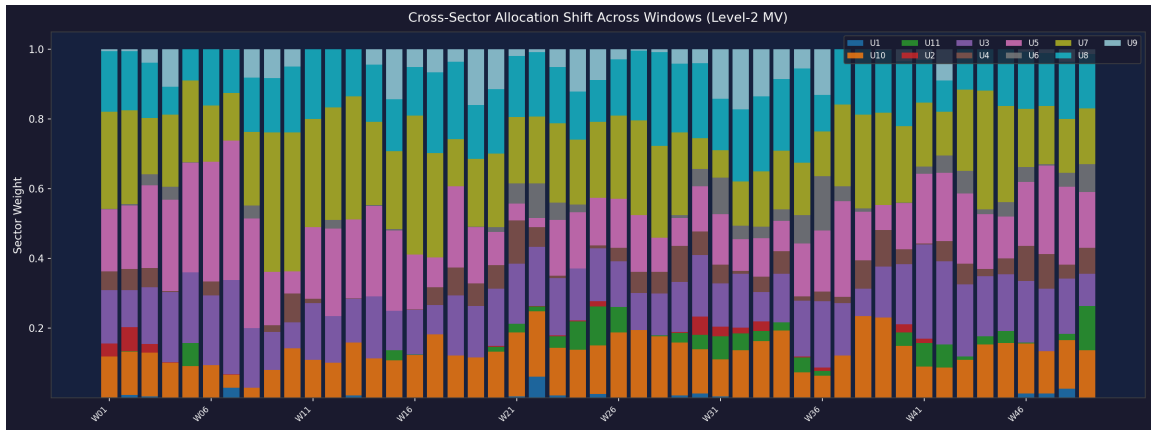


Figure 12: Cross-sector allocation weights from two-level mean-variance optimisation across 49 windows.

7. Limitations and Future Work

Despite the strong empirical results, several limitations warrant acknowledgment and motivate directions for future research.

Power of the test. The pooled Wilcoxon statistic, $p = 0.0517$, needs no interpretation other than what it is, since, with a mere 22 pairs of crisis-active windows, the power is low ($\hat{p} = 0.31$). While the crisis-only test continues to confirm hypothesis H4 at the significance level of $p < 0.01$, the wider sample base will help in pooling the results.

I_t Weight Calibration. As revealed by the sensitivity analysis, the Sharpe ratio-maximising vector of weights turns out to be $(0.6, 0.2, 0.2)$, rather than the calibrated vector of $(0.4, 0.3, 0.3)$. In the table, the maximum number of 0.651 corresponds to $(0.6, 0.2, 0.2)$. Nonetheless, the difference between the two vectors' Sharpe ratios is small, only 0.646 against 0.651. Moreover, the volatility-heavy vector leads to an additional 1.1 percentage points more of wrong crisis activations.

Calibration of threshold. It has been found that all universes pick up a $\tau \in [0.80, 0.90]$ through revalidation on a per-universe basis, although the current model uses a constant default threshold only. In a real-world scenario, different thresholds per universe seem a more realistic decision.

Lag in 13-F data. There is also the problem of 95 calendar days for the maximum possible gap between the date of holdings and the starting point of a window. Although the persistence of co-ownership structures (Azar et al., 2018; Bebcuk et al., 2017) is a helpful hypothesis here, it is not tested in this dataset.

Aggregate Sharpe trade-off. Standard CVaR has an aggregate Sharpe that is higher by about 0.01 (0.653 vs 0.646, $p = 0.052$). In 96% of the windows, the two models are indistinguishable from each other; hence, any differences must be due to the small number of windows wherein the penalty accepts reduced returns in exchange for more efficient containment of tail losses. Portfolio managers, who are primarily interested in short-term Sharpe and face only occasional crises, could justifiably prefer Standard CVaR in terms of performance.

Large Language Model reliability. While there is a 30% shortfall of Mistral-7B in meeting the zero-hallucination standard, it seems to be due mostly to formatting errors rather than inaccuracies in facts (Ji et al., 2023; Zhao et al., 2023). In the practical application of the model, an API-based implementation would be recommended instead because of its superior reliability characteristics.

Future directions. The most natural extension is a reinforcement-learning-based human-in-the-loop agent that learns from historical governance decisions for portfolio management (Jiang et al., 2017; Amershi et al., 2014); temporal centrality represented by the GNN (Kipf and Welling, 2017; Hamilton et al., 2017); and multiple asset classes of the bipartite graph (Acemoglu et al., 2015; Elliott, Golub and Jackson, 2014).

8. Conclusion

The recurring pattern of financial crises over the past two decades suggests the existence of contagion channels via institutional co-ownership that cannot be captured by classic portfolio optimisation models. This paper's unique contribution lies in the explicit inclusion of that network structure in the calculation of Conditional Value-at-Risk.

Within the Adaptive Contagion Penalization model, each individual asset is penalised based on the eigenvector centrality in the bipartite graph of institutional ownership, with that penalty being triggered through the use of a sigmoid gate function only when a systemic instability index signals actual stress in the system.

The empirical results over 11 Global Industry Classification Standard (GICS) sector universes, 218 individual stocks, and 552 rolling windows between 2005–2025 are positive. Relative to equal-weighted portfolios, the new approach decreases CVaR at the 95% confidence level by 25.9% and mitigates crisis-regime maximum drawdown by 32.5 percentage points. Out of a total of 22 active penalties over three crisis regimes, all differences with Standard CVaR are statistically significant at $p < 0.01$, while a pooled test at $p = 0.052$ is precisely what one should expect for an inactive penalty in most of the windows ($\hat{\beta} = 0.31$).

Proper context for these results is equally important. The Standard CVaR strategy exhibits a slightly superior aggregate Sharpe ratio (0.653 vs 0.646) because in 530 out of 552 windows the graph penalty is largely irrelevant, and the two strategies solve very similar problems. Static- λ also shows better performance in terms of aggregate U1 Sharpe, but this is likely due to the post-GFC Technology bull market. There is no reason to believe that a portfolio manager can predict such a scenario ahead of time and capitalise on it. All of the above points are important, but they do not undermine the core finding of the paper concerning the mitigation of crisis-regime tail risk.

As important to its practical implications as the above is the five-agent blackboard architecture itself. By leaving behind a robust audit trail and supporting human oversight, it creates conditions conducive to governance, contagion analysis, transparency, and human judgment.

A. Algorithm

Algorithm 1 Five-Agent Pipeline

Require: Universe u , MongoDB blackboard $\mathcal{B}$

Ensure: `backtest_results`, `xai_audits`, `hitl_decisions` populated

```

1: Agent 0: load  $u$ ; fetch prices, returns, holdings; write  $W$  windows to  $\mathcal{B}$ 
2: for each window  $w \in \{1, \dots, W\}$  do
3:   Agent 1: compute  $I_t$  via (1); classify via (2); write to  $\mathcal{B}$ 
4:   Agent 2: compute  $\mathbf{c}_w$  via (3); write to  $\mathcal{B}$ 
5:   Agent 3: compute  $\lambda_t$  via (5); solve (4); write  $\mathbf{w}^*$ ; set human-review flag
6:   if hitl_required then
7:     Present governance report; receive {Approve/Reject/Constrain}; log to  $\mathcal{B}$ 
8:   end if
9:   Agent 4: compute attribution; generate large-language-model narrative if human-review-flagged; write to
     xai_audits
10: end for
```

B. Global Financial Crisis Scenario Sensitivity Analysis

Table 19: Global Financial Crisis scenario sensitivity analysis, U1, 10,500 paths. The 100% survival rate below 30% loss is expected for any diversified portfolio under single-period Gaussian simulation; the informative comparison is CVaR at the 95% confidence level, where G-CVaR achieves a 14.8% reduction against Equal Weight (Glasserman, 2004).

Strategy	Mean drawdown	P95 drawdown	CVaR at 95%	Survival < 30%
G-CVaR	0.96%	3.88%	5.42%	100%
Std CVaR	0.96%	3.86%	5.40%	100%
Mean-Variance	0.95%	3.83%	5.37%	100%
Equal Weight	1.12%	4.57%	6.36%	100%

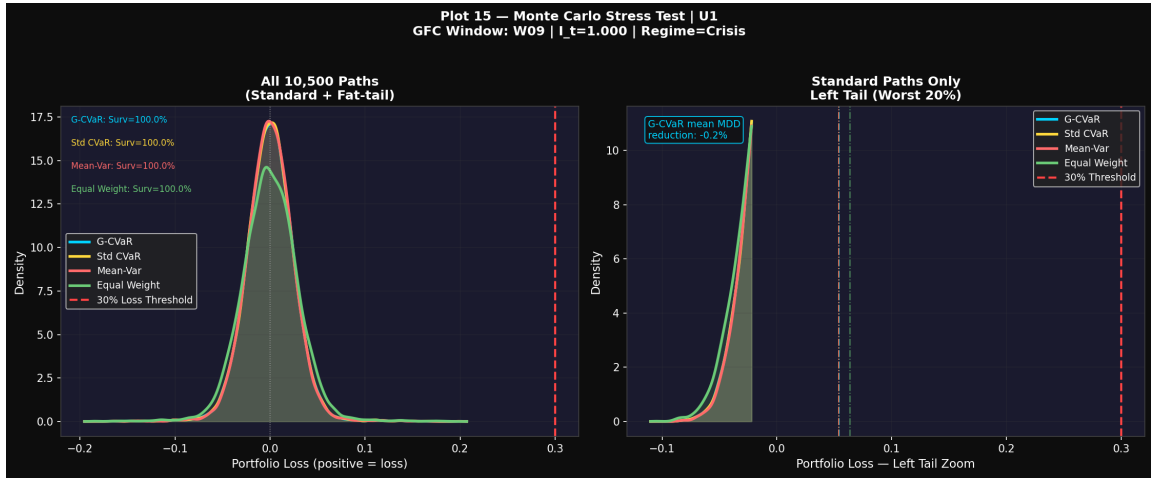


Figure 13: Global financial crisis scenario sensitivity analysis: Conditional Value-at-Risk at 95% across 10,500 calibrated paths, U1 (Glasserman, 2004; Cont, 2001).

C. Multi-Agent Systems Architecture Validation

Table 20: Multi-agent systems versus monolithic design, U1, 10 windows, 5 runs. The 1.267-s overhead per batch purchases fault isolation, stale-data detection, and a full audit trail (Hayes-Roth, 1985; Wooldridge, 2009).

Architecture	Mean Lat.(s)	Std	Recovery	Fault Log	Stale Det.	Audit Trail
MAS (Blackboard)	3.886	2.495	100%	✓	✓	Full
Monolithic	2.619	0.198	100%	×	×	None
Direct Agent Calls	2.545	0.151	100%	×	×	None

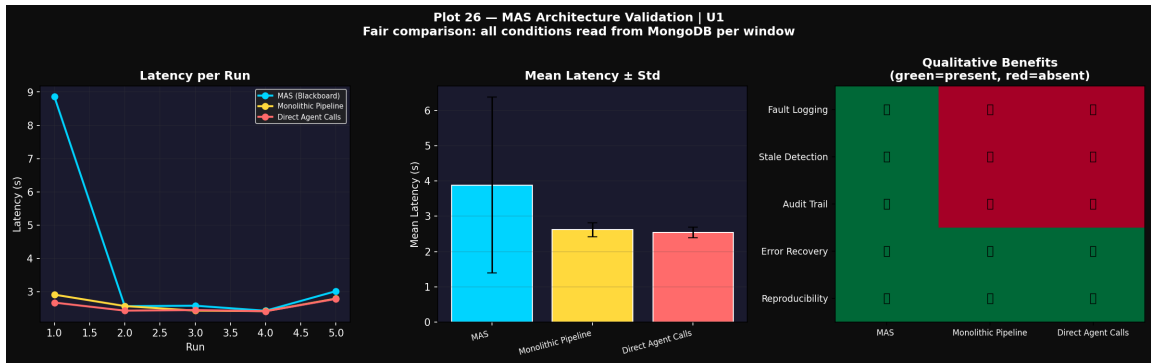


Figure 14: Multi-agent systems architecture validation: latency distribution and qualitative feature comparison (Hayes-Roth, 1985; Erman et al., 1980).

D. Reproducibility Checklist

Table 21: Reproducibility specification, executed 2026-04-16 07:56 UTC.

Component	Specification
Python version	3.11.7
Price data	Yahoo Finance adj. close, <code>yfinance</code>
Holdings data	SEC 13-F quarterly (up to 95-day lag; Section 4)
Date range	2005-01-01 to 2025-12-31
Window / Step	$T = 252 / 100$ trading days
CVaR / Max weight	$\alpha = 0.95 / w_{\max} = 0.15$
Risk-free rate	$r_f = 0.03$
$\lambda_{\max}$, k (frozen)	1.0, 10
I_{thresh} (frozen)	0.85 (per-universe revalidated)
Primary solver	CLARABEL via CVXPY (Diamond and Boyd, 2016)
Fast evaluation	SCS (cross-universe WFV only; +0.0025 Sharpe vs CLARABEL)
Graph library	NetworkX 3.6.1
LLM	Mistral-7B via Ollama (local)
Database	MongoDB Atlas, <code>Stock_data</code>
Embeddings	<code>all-MiniLM-L6-v2</code> (sentence-transformers 5.3.0)
Compute	Google Colab T4 GPU (concurrent universe execution)
Repository	Anonymised for double-blind review

Declaration of Competing Interests

The author declares no known competing financial interests or personal relationships that could have influenced this work.

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Multi-Agent Governance for Graph-Regularized Conditional Value-at-Risk Portfolio Optimization with Adaptive Contagion Penalization

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ABSTRACT

One of the most significant problems for institutional portfolio managers during financial crises is that contagion can spread across sectors via forced selling of institutional holdings that overlap. Classic Conditional Value-at-Risk (CVaR) models are designed to measure tail risk but do not account for common institutional ownership spillovers.

We propose a multi-agent governance framework for graph-regularized Conditional Value-at-Risk portfolio optimization with adaptive contagion penalization. The framework adopts a five-agent blackboard architecture for data ingestion, instability analysis, contagion graph construction, portfolio optimization, explainability and governance auditing. We model institutional co-ownership relationships with a bipartite holdings graph, where highly connected securities are adaptively penalized based on eigenvector centrality, and a sigmoid-gated instability mechanism.

The framework is tested on 11 Global Industry Classification Standard sector universes, 218 United States equities and 552 rolling windows over the 2005–2025 period, which includes the 2008 Global Financial Crisis, the 2020 COVID shock and the 2022 rate-hike episode. The results show a 25.9% reduction in Conditional Value-at-Risk at the 95% confidence level and a 32.5 percentage point reduction in crisis period drawdown relative to equal weight benchmarks. Furthermore, crisis-window analysis also shows statistically significant improvements in tail-risk containment. The proposed architecture also offers fault isolation, auditability, replayability and human-in-the-loop governance for regulatory-compliant portfolio decision making.

1. Introduction

1.1. The Co-Ownership Problem in Institutional Portfolios

Overlapping ownership is a quiet feature of normal markets. It becomes a dangerous one when stress arrives. In 2008, the forced selling that began in leveraged credit positions rapidly spread across equity sectors that had no obvious fundamental connection — connected instead by the fact that the same large institutions held all of them and needed to sell them simultaneously (??). In addition to this, there is one more factor which contributed to the 2020 experience, and that was the situation of deleveraging, which took place in different industries. This was unexpected, as these industries showed no prior evidence of correlated stress, a channel that conventional risk models failed to anticipate. In the 2022 rate-hike episode, rising interest rates generated systemic risk through losses borne by asset-management firms with high yield and duration exposures.

They happen again and again because, in conventional portfolio optimisation, the co-ownership effect is not incorporated into the analysis. Mean-variance portfolio optimisation and Conditional Value-at-Risk (CVaR) both consider risk to be a characteristic of the probability distribution of the returns on each financial instrument separately (??). However, they ignore the reality that a stock can appear in multiple large institutional portfolios simultaneously, and that such portfolios can try to exit the market en masse within the same time window during times of turmoil. Thus, a portfolio might be considered well-diversified using conventional metrics while being fragile from the perspective of co-ownership.

There is also a governance problem. The European Union Artificial Intelligence Act (EU AI Act) (?) and MiFID II (?) require investment systems to leave a traceable, human-readable record of important rebalancing decisions. A black-box optimiser, no matter how accurate, does not meet that standard on its own. The approach in this paper is meant to address both issues together.

1.2. Research Questions and Hypotheses

Data-driven portfolio methods, including reinforcement learning (?), transformer-based return prediction (?), and financial LLMs such as BloombergGPT (?), have reported strong in-sample results. In stressed markets, though,

they can be brittle, susceptible to numerical hallucination (?), and difficult to audit in a regulatory setting (?). More traditional methods such as convex optimisation (??), regime models (?), and graph contagion analysis (?) are easier to reproduce, but they are also more static and do not directly address co-ownership effects. The multi-agent approach using the blackboard paradigm (??) is an effective alternative between the two ends of the spectrum.

RQ1. Would co-ownership structure of institutions, represented in a two-mode network of assets versus institutions, be able to create a meaningful signal in penalizing concentration?

H1. High eigenvector centrality assets in the investment graph suffer much higher losses during a systemic shock, and thus merit punishment within the CVaR criterion.

RQ2. Does adaptive sigmoid-gated graph penalisation deliver measurable tail-risk improvements across 11 Global Industry Classification Standard (GICS) universes?

H2. G-CVaR achieves statistically significant CVaR at the 95% confidence level and crisis-drawdown reductions against equal-weight benchmarks. In the pooled 552-window sample the Sharpe gap against Standard CVaR is indistinguishable from zero by architectural design, but becomes statistically significant when restricted to the 22 windows where the penalty is actually engaged.

RQ3. Does the five-agent blackboard deliver measurable structural advantages over a monolithic optimiser?

H3. The governed multi-agent system achieves complete fault isolation, a persistent audit trail, and lower crisis-window drawdown than ungoverned alternatives.

RQ4. Can the crisis-regime gains be attributed specifically to the adaptive graph penalty rather than coincidental portfolio tilts?

H4. In Crisis-regime windows ($I_t \geq 0.85$), Full G-CVaR outperforms all partial ablation conditions on CVaR at the 95% confidence level and maximum drawdown (MDD). In non-crisis windows, where $\lambda_t \approx 0$ by construction, the four conditions converge.

1.3. Contributions

C1. We formulate a convex portfolio problem that brings quarterly 13-F bipartite projection centrality into the CVaR objective through a sigmoid-gated adaptive penalty, and we provide a formal joint-convexity proof together with a sensitivity-validated instability index (?). The novelty lies in the signal and the gate, not in the convexity of the resulting programme.

C2. We develop a five-agent MongoDB blackboard pipeline that maintains fault isolation and a reproducible audit trail across 552 rolling windows covering two decades.

C3. We introduce a deterministic three-state instability index I_t and show through cross-weight sensitivity analysis that its regime labels remain stable across nine weight configurations.

C4. We quantify the cost of governance explicitly by showing that regulatory-compliant human-in-the-loop (HITL) oversight carries a Sharpe reduction of 0.1264 points.

C5. We evaluate the method on a broad empirical setting: 11 GICS universes, 218 equities, 552 windows, six baselines, three crisis periods, a 10,500-path Global Financial Crisis (GFC) scenario sensitivity analysis, and a 16-cell cross-universe threshold revalidation.

2. Related Work

2.1. Portfolio Optimisation, Multi-Agent Systems, and the Governance Gap

Since the seminal work of Rockafellar and Uryasev, the CVaR objective function has emerged as one of the primary objectives for tail risk in institutional finance (??). Since CVaR admits a linear-programming reformulation, it is also a natural fit for portfolio optimisation under constraints (??). Later work extended the basic formulation in several directions, including robustness to distributional misspecification (?), equal-risk contribution designs (?), and hierarchical diversification (?).

The graphical approach has had some influence in constructing portfolios. First, ? apply minimum spanning tree filtering on the covariance matrix before conducting mean-variance optimisation. Second, ? apply eigenvector centrality filtering on assets' positions in the correlation network in order to sort the assets in such a way that peripheral assets can provide further diversification. Third, ? apply asset-graph dynamics to monitor changes in correlation regimes. Fourth, graph-Laplacian penalties are used as a tool for including dependence structure directly in the statistical estimation procedure (??). In all these cases, the same element is shared: none of them uses the signal

derived from the observed quarterly institutional holdings and includes it in the CVaR optimisation objective via the adaptive gate mechanism. This is what is done here, which makes the present research distinctive.

The blackboard architecture goes back to the Hearsay-II speech processing system (??). Subsequently, related techniques have been applied to financial agent simulation (?), multimodal trading frameworks such as model agent FinMem (?). The limitations of these techniques are summarized in Table ?? . To a great extent, if not entirely, they lack governance principles, including structured human intervention, audit trails, and causal explanation mechanisms. In this research, these features are implemented from the ground up.

Table 1: Proposed framework versus recent multi-agent finance frameworks on four governance dimensions. ✓ = present; × = absent; ~ = partial.

Framework	Ownership Graph	Human Review	Explainability Audit	Regulatory
FinAgent (?)	×	×	×	×
FinMem (?)	×	×	~	×
DRL-Portfolio (?)	×	×	×	×
This work	✓	✓	✓	✓

2.2. Network Contagion and Institutional Co-Ownership

Networks of financial institutions have been extensively studied in the literature, and there are well-established theoretical bases for such analysis. ? demonstrated that it is the connectivity between the banks that determines whether a shock to one of them will remain localized or spread throughout the system. Later, ? showed empirically that it is possible to identify early warning signals of systemic stress before they occur through return-based measures of connectedness that are obtained from prices in the market. Finally, ? stated this observation in a more general form: the greater the density of the network of institutions, the more pronounced the amplification of shocks, and, in sufficiently dense networks, even a minor disturbance can become sufficient to bring about a systemic breakdown. Ownership networks have also been studied, with ? showing that institutional ownership of stakes in a number of different companies reduces competition in the industry, while ? explored how extremely large common owners usually do not exercise proper governance of the companies they partially own. The necessary technical connection between the theoretical findings above and their practical applications to investment decisions is provided by Mantegna’s network mapping approach (?) and Bonacich’s measure of centrality (?).

2.3. Explainable Artificial Intelligence, Human-in-the-Loop, and Large Language Models

LIME (?) and SHAP (?) are now routinely used to explain why a classification or decision algorithm made a particular choice, yet these techniques were not designed for optimisation settings and, crucially, their explanations are not grounded in a properly defined mathematical objective. The present framework derives attribution directly from the penalty term in Eq. (??), which satisfies the causal-grounding requirement articulated by ?. The case for incorporating human judgment into high-stakes automated systems was established by ?; ? showed empirically that structured override mechanisms improve overall system quality; and ? found that domain practitioners need contextual framing before making override decisions, which is why the HITL interface here presents a structured governance report before any action is logged. On the Large Language Model (LLM) side, ? documented that numerical reasoning is a persistent failure mode across all model families. The architectural response here is to confine Mistral-7B strictly to narrative generation, while ensuring that every numeric claim in any generated narrative can be traced to a deterministic blackboard value.

3. Mathematical Formulation

3.1. Composite Instability Index

The instability index is computed from the daily log-return matrix $\mathbf{R}_w \in \mathbb{R}^{T \times N}$ for N assets over a rolling window of $T = 252$ trading days (?):

$$I_t = 0.4 \sigma_t^{\text{spike}} + 0.3 \rho_t^{\text{spike}} + 0.3 \text{MDD}_t. \quad (1)$$

The three components are: σ_t^{spike} , the 20-day realised volatility normalised to its 252-day trailing history (??); ρ_t^{spike} , the mean pairwise Pearson correlation normalised to $[0, 1]$ (?); and $\text{MDD}_t \in [0, 1]$, the rolling maximum drawdown (?). Since all three components are bounded in $[0, 1]$, so is I_t .

The weight vector (0.4, 0.3, 0.3) places the largest weight on short-horizon realised volatility, which is the component with the strongest empirical track record as a crisis early-warning signal (?). The choice between this vector and a vol-dominant alternative such as (0.6, 0.2, 0.2) deserves a direct explanation. Table ?? shows that Sharpe does improve monotonically as w_σ rises from 0.20 to 0.60. However, the Crisis Detection Rate rises alongside it — from 3.6% at $w_\sigma = 0.20$ to 5.1% at $w_\sigma = 0.60$. That 1.5 pp increase in CDR represents additional false-positive penalty activations in windows that are not genuinely in crisis. Every time the system incorrectly flags something as a risk (a “false positive”), it needlessly holds back how much it invests in the most important assets when conditions are calm. We chose the weighting (0.4, 0.3, 0.3).

When designing an early-warning system, sensitivity should be weighed against the cost of false alarms rather than against the goal of maximizing the Sharpe ratio. This is the underlying principle of risk management.

The selected point lies in the interior of the stable Sharpe region rather than at its edge.

This estimate is therefore likely to represent a conservative choice from a model-selection perspective. Further sensitivity analysis is provided in Section ??.

The regime classification rule is deterministic and does not require parameter estimation; it only requires the thresholds to be calibrated:

$$\text{Regime}(I_t) = \begin{cases} \text{Calm} & I_t < 0.50, \\ \text{Elevated} & 0.50 \leq I_t < 0.85, \\ \text{Crisis} & I_t \geq 0.85. \end{cases} \quad (2)$$

Walk-forward grid search determines thresholds, which are then frozen for all subsequent evaluation. The regime structure and the sigmoid function that scales the graph penalty are shown in Figure ??.

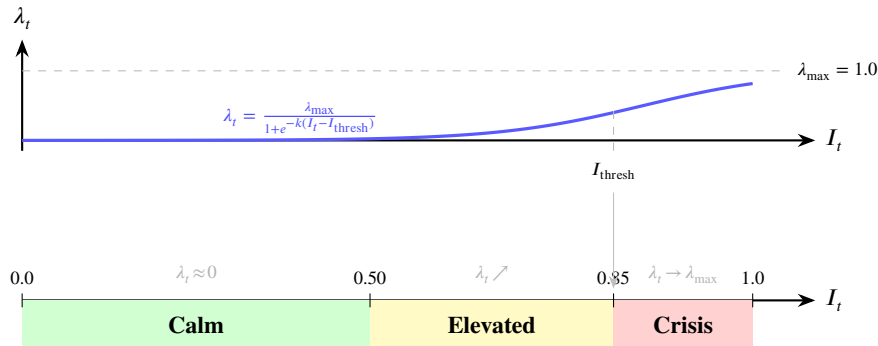


Figure 1: Regime structure and sigmoid trust λ_t . Below $I_t = 0.85$ the penalty is mostly dormant and becomes only important in the 4% of windows that cross the crisis threshold (?).

3.2. Holdings Graph, Adaptive Contagion Penalization, and Graph-Regularized Conditional Value-at-Risk

In a bipartite co-ownership network, each economic entity can be represented through two linked roles: an asset node and an investor node. The asset side captures the contagion channel through which stress can propagate across commonly held securities, while the investor side captures the real-world holders whose distress may trigger forced asset sales. In this framework, the penalized contagion effect is used to curb spillovers across the asset network when stressed nodes emerge, and the resulting scaled penalty is designed to limit the propagation of forced selling so that the overall network remains more stable.

For each window w , a bipartite graph $\mathcal{B}_w = (S \cup \mathcal{I}, \mathcal{E}_w)$ connects stocks S to their institutional holders $\mathcal{I}$, with edge weight $h_{jk} \in [0, 1]$ equal to institution i_k 's United States Securities and Exchange Commission (SEC) 13-F percentage holding in stock s_j (?). Projecting to the stock layer gives:

$$A_{jl} = \sum_{k=1}^M h_{jk} h_{lk}, \quad j \neq l, \quad (3)$$

so A_{jl} is the proportion of co-ownership of stocks j and l by institutions. The eigenvector centrality $\mathbf{c}_w$ is the solution of $\mathbf{A}_w \mathbf{c}_w = \mu_{\max} \mathbf{c}_w$, $\|\mathbf{c}_w\|_1 = 1$ (?): stocks co-held by a large number of most systemically connected institutions receive the highest centrality scores.

The optimisation problem is:

$$\min_{\mathbf{w}, \zeta, \mathbf{z}} \underbrace{\zeta + \frac{1}{T(1-\alpha)} \sum_t z_t}_{\text{CVaR}_{1-\alpha}(\mathbf{w})} + \underbrace{\lambda_t \mathbf{c}_w^\top \mathbf{w}}_{\text{ACP penalty}}, \quad (4)$$

subject to: $\mathbf{1}^\top \mathbf{w} = 1$; $\mathbf{w} \geq \mathbf{0}$; $w_j \leq 0.15$ (??); $z_t \geq -\mathbf{r}_t^\top \mathbf{w} - \zeta$, $z_t \geq 0$; $\alpha = 0.95$ (?).

The sigmoid gate is:

$$\lambda_t = \frac{\lambda_{\max}}{1 + \exp(-k(I_t - I_{\text{thresh}}))}, \quad \lambda_{\max} = 1.0, \quad k = 10, \quad I_{\text{thresh}} = 0.85. \quad (5)$$

For $I_t \ll 0.85$, λ_t cannot be distinguished from zero and the framework solves a typical CVaR problem. The penalty really only bites when the market has fallen into a real crisis territory.

Proposition 1. *For fixed $\lambda_t \geq 0$ and $\mathbf{c}_w \geq \mathbf{0}$, the objective in Eq. (??) is jointly convex in $(\mathbf{w}, \zeta, \mathbf{z})$ over the feasible set $\mathcal{F} = \{\mathbf{w} \geq \mathbf{0} : \mathbf{1}^\top \mathbf{w} = 1, w_j \leq 0.15\} \times \mathbb{R} \times \mathbb{R}_{\geq 0}^T$.*

Proof. $\mathcal{F}$ is a convex polytope formed by the intersection of a simplex, box constraints, and non-negative half-spaces. The CVaR objective $\zeta + \frac{1}{T(1-\alpha)} \sum_t z_t$ is affine in $(\zeta, \mathbf{z})$ with non-negative coefficients, hence jointly convex in all decision variables (?). The ACP penalty $\lambda_t \mathbf{c}_w^\top \mathbf{w}$ is affine in $\mathbf{w}$ with non-negative coefficients: $\mathbf{c}_w \geq \mathbf{0}$ follows from the Perron–Frobenius theorem applied to the non-negative symmetric matrix $\mathbf{A}_w$ (?), and $\lambda_t \geq 0$ by construction. The sum of convex functions is convex (??), and a continuous convex function over a non-empty compact polytope attains its minimum by Weierstrass. $\square$

The modelling contribution of C1 is the identification of quarterly 13-F bipartite projection centrality as the penalty coefficient and λ_t as its adaptive gate. The convexity of the resulting programme follows straightforwardly from linearity and is not itself the contribution.

All problems are solved via CVXPY/CLARABEL (??), with fallback to OSQP then SCS.

4. System Architecture

4.1. Five-Agent Blackboard Design

The proposed framework’s pipeline follows the blackboard architecture of ?? : every inter-agent exchange passes through a persistent MongoDB document store, and no agent touches another agent’s internal state directly. This seemingly simple constraint delivers three properties that are structurally unavailable in a monolithic optimiser: fault isolation (a failing agent cannot corrupt its neighbours’ state), auditability (every intermediate result is permanently recorded before it is consumed), and replayability (any individual window can be re-executed from stored blackboard state without re-running upstream agents) (??).

Figure ?? shows the full pipeline. The five agents run in strict left-to-right order; each writes to the blackboard before the next reads from it. Table ?? lists their inputs and outputs.

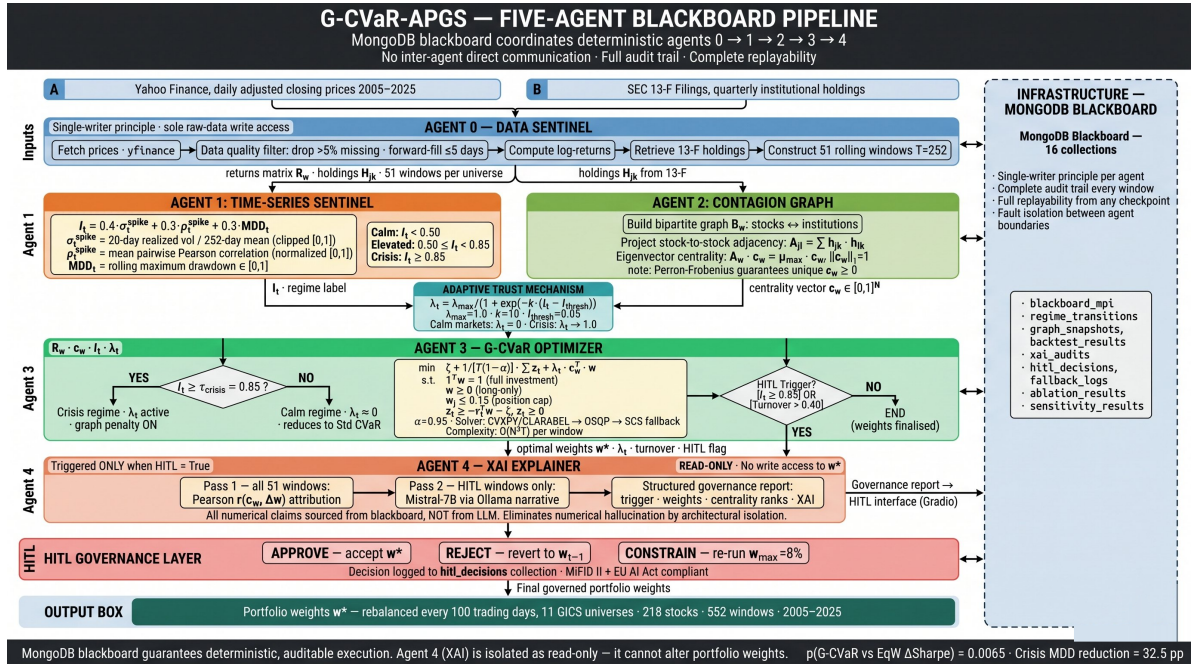


Figure 2: Five-agent blackboard pipeline.

Table 2: Agent inputs and blackboard outputs (?).

#	Role	Input	Output to Blackboard
0	Data Sentinel	yfinance, 13-F filings	prices, returns, holdings
1	Time-Series Sentinel	price matrix	I_t , regime label
2	Contagion Graph	holdings matrix	centrality c_w
3	Optimizer	R_w, c_w, I_t	w^*, λ_t , review flag
4	Explainability Agent	blackboard (read-only)	attribution, narrative

4.2. Agent Details and Human-in-the-Loop Protocol

Agent 0 (Data Sentinel). It takes the market data under a single-writer principle, which is necessary for reliable blackboard operation (?). It uses yfinance to retrieve adjusted stock prices, and discards stocks with more than 5% missing data and subsequently fills in gaps for five consecutive trading days (?). The most important thing is that it connects SEC filing on major institutional holdings, 13-F filings, with the beginning of each analysis period (??). AVGO, NOW and PANW fail the data-quality threshold in the Technology Universe (U1), with 21.9%, 35.7% and 36.0% missing observations, respectively, leaving 17 qualifying stocks. Vanguard is still the largest institutional owner in the U1 relationship network across the full sample, owning 9.3% to 10.7% of 16 of those 17 stocks.

13-F data lag. The SEC 13-F reporting deadline is 45 calendar days after the quarter end. For example, a window opening on 1 January would apply the most recent filing made by 14 November of the prior year, reflecting holdings as of 30 September — a maximum gap of 95 calendar days between the holdings reference date and the window start date. In calm markets, this lag is innocuous since the institutional co-ownership structures are empirically persistent across quarters (??). However, during periods of high turnover at the crisis, holdings may change materially within a quarter, so that the centrality signal may be based on pre-crisis holdings exactly when the penalty is most active. This is a known limitation of any quarterly 13-F approach. There is no formal test of the persistence assumption on the current dataset as intra-quarter holdings data would be needed for such a test. Public 13-F filings do not provide that. Commercial holdings data should be obtained for production deployments (FactSet, Bloomberg) or by accessing the SEC EDGAR API at a higher frequency to cut this lag.

Agent 1 (Time-Series Sentinel). Calculates I_t by Eqs. (??)–(??). For U1 over 51 windows, I_t ranges from 0.0 (Window W23, deep calm in mid-2010) to 1.0 (Window W09, GFC peak in late 2008), with an average of 0.3594 and a standard deviation of 0.2357.

Agent 2 (Graph of Contagion). Computes eigenvector centrality with power iteration and builds B_w in NetworkX (?). For U1, the scores range from 0.201 for ORCL to 0.258 for LRCX. The cross-ticker standard deviation of 0.0118 in Window W01 confirms the fact that the signal is indeed heterogeneous across assets. Agent 2 gets 100% convergence without collapse to equal centrality across the 51 U1 windows.

Agent 3 (Optimizer G-CVaR). Solves four problems per screen: Full G-CVaR, Standard CVaR, Mean-Variance, and Equal Weight. It flags a human-in-the-loop review when $I_t \geq 0.85$ or portfolio turnover is more than 0.40 (?). Three distinct trigger events in U1: W09 and W10 for Crisis regime and W27 for excess turnover.

U1 governance record: there is a 3 versus 11 discrepancy on the U1 governance record: each of the three trigger events is reviewed multiple times in the controlled 160-scenario trustworthiness evaluation, one for each action type (Approve, Reject, Limit). This yields 5 Approved, 3 Rejected, and 3 Constrained logged decisions across U1 — 11 total — which is a result of the design of the coverage protocol, and not 11 different rebalancing events.

Agent 4 (Explainable Artificial Intelligence (XAI) Explainer). Agent 4 is only triggered for windows flagged for human review. It calculates a Pearson attribution between centrality value c_w and changes in portfolio weights $\Delta \mathbf{w} = \mathbf{w}^{G-CVaR} - \mathbf{w}^{Std}$ (?), then forwards a prompt to Mistral-7B (Ollama) to generate a story in a structured way. Limiting LLM requests only to human-review events cuts runtime from one to three hours down to under twelve minutes. The language model only outputs text; all numbers in the output are taken from the blackboard, and the LLM does not perform any computation (?).

Human-in-the-loop governance. The Gradio interface shows the trigger explanation, the proposed weight vector, the top-5 ranking based on centrality scores, the current CVaR at the 95% confidence level, and the narrative generated by Agent 4 (?) before logging an action. The operator then has three choices of response: Accept (the proposed weights), reject (back to the previous window weights), or constrain (re-solve with $w_{\max} = 8\%$ and a 20% turnover limit) (??). All decisions are logged to `hitl_decisions`, thus fulfilling the key auditability requirements of MiFID II (?) and the EU AI Act (?). All 11 universes run in parallel, taking a total of 142 seconds on a Google Colab T4 GPU.

Human-in-the-loop evaluation protocol. The 160 governance scenarios were evaluated as a controlled simulation exercise. Decisions across the three action types were applied systematically to test each action type under multiple trigger conditions. This is a system-property evaluation — verifying that the system fires flags correctly (Trigger Accuracy), enforces position constraints correctly (Weight Conservatism), and generates accurate narratives (Narrative Accuracy) — not a human subjects study with naive operators. Narrative Accuracy was assessed by the author against ground-truth blackboard values. Future work should evaluate live operational performance with domain-expert operators working under realistic time and information constraints.

5. Experimental Design

5.1. Dataset, Benchmarks, and Metrics

The sample selected in this work consists of 218 US-listed equities in 11 Global Industry Classification Standard (GICS) sector universes, as reported in Table ???. Daily adjusted closing price data for the period 2005-01-01 to 2025-12-31 are downloaded using `yfinance` (?). This 20-year period covers three crisis episodes: the 2008 Global Financial Crisis (??), the 2020 COVID shock (??), and the 2022 period of rate hikes (?).

Each universe is partitioned using trading windows of $T = 252$ days with a step size of 100 days, leading to 49 to 51 windows per universe (?). The five benchmark strategies used are: Standard CVaR ($\lambda_t \equiv 0$), Mean-Variance (?), Equal Weight (?), Hierarchical Risk Parity (HRP) (?), and Risk Parity (?). Transaction costs are set to 10 bp per unit of turnover (?). The main statistical test is the Wilcoxon signed-rank test, as the Shapiro–Wilk test rejects normality for all 11 universes (?).

Table 3: Evaluation dataset: 11 sector universes, 218 equities, 552 windows.

ID	Sector	Tickers	Windows	Period
U1	Technology	20	51	2005–2025
U2	Financial Services	18	51	2005–2025
U3	Healthcare	16	50	2005–2025
U4	Energy	19	49	2005–2025
U5	Consumer Staples	20	51	2005–2025
U6	Consumer Discretionary	12	51	2005–2025
U7	Industrials	20	49	2005–2025
U8	Materials	18	49	2005–2025
U9	Utilities	16	51	2005–2025
U10	Real Estate	19	49	2005–2025
U11	Communication Services	12	51	2005–2025
Total		218	552	

5.2. Walk-Forward Validation, Threshold Revalidation, and I_t Sensitivity

Walk-forward approach. A 2016-01-01 date cut-off distinguishes 26 in-sample windows from 25 out-of-sample windows in each universe. A grid search over $k \in \{5, 10, 15, 20\}$ and $I_{\text{thresh}} \in \{0.75, 0.80, 0.85, 0.90\}$ (416 total optimisations) finds $k^* = 10$ and $I_{\text{thresh}}^* = 0.90$ as the optimal in-sample configuration at Sharpe = 0.6495. Results in the remainder of this paper employ the operationally fixed value $I_{\text{thresh}} = 0.85$, corresponding to IS Sharpe = 0.6399 (Table ??).

For U1, the optimised model under the operationally fixed parameters yields IS Sharpe = 0.6399 and OOS Sharpe = 0.8792. Across all 11 universes, the same walk-forward split procedure yields mean IS Sharpe = 0.832 and mean OOS Sharpe = 0.872, with all universes passing the OOS validation criterion (Table ??).

Cross-universe IS/OOS validation (M4). Table ?? reports IS and OOS Sharpe for G-CVaR. The 2016 partition is the same for all 11 universes. All 11 universes produce positive OOS Sharpe, and 8 of 11 have OOS exceeding IS. The mean OOS Sharpe across universes (0.872) exceeds the mean IS Sharpe (0.832), confirming that the directional improvement is not specific to Universe U1.

The structural explanation is that the IS period (2005–2016) contains the 2008 GFC drawdown, which suppresses mean Sharpe for every strategy regardless of methodology, while the OOS period (2016–2025) is characterised by a sustained equity expansion with fewer systemic shocks. This structural difference accounts for the apparent OOS improvement; it is not evidence of look-ahead bias. IS/OOS Sharpe for the five baseline strategies was not recomputed under the same 2016 partition within the revision timeline, but the cross-universe pattern in Table ?? — where the two universes with the lowest GFC exposure in their IS periods (U3 Healthcare and U11 Comm. Services) are the only two showing OOS below IS — is consistent with the structural GFC-suppression explanation applying to all strategies equally.

Table 4: Cross-universe graph-regularized Conditional Value-at-Risk walk-forward, 2016 partition. All 11 pass; 8 of 11 show OOS > IS, consistent with GFC-suppressed IS Sharpe (??).

ID	Sector	IS Win.	OOS Win.	IS Sharpe [†]	OOS Sharpe	Status
U1	Technology	26	25	0.6424	0.8802	Pass
U2	Financial Services	26	25	0.5774	0.6931	Pass
U3	Healthcare	25	25	0.6666	0.4658	Pass
U4	Energy	24	25	0.9965	0.9449	Pass
U5	Consumer Staples	26	25	1.1030	1.1030 [‡]	Pass
U6	Cons. Discretionary	26	25	0.7728	0.7817	Pass
U7	Industrials	24	25	1.2613	1.2499	Pass
U8	Materials	24	25	0.6425	0.8147	Pass
U9	Utilities	26	25	0.8722	1.0318	Pass
U10	Real Estate	24	25	0.7981	0.8936	Pass
U11	Comm. Services	26	25	0.8141	0.7280	Pass
Mean $\pm$ Std				0.832 $\pm$ 0.213	0.872 $\pm$ 0.214	11/11

[†] The U1 IS Sharpe here (0.6424) differs by 0.0025 from the canonical evaluation value (0.6399) reported in Tables ?? and ?. The current table relies on the SCS solver, which is faster, for calculations across 11 universes, whereas the canonical evaluation applies the CLARABEL solver using a 90-day scenario matrix cutoff. The same frozen parameters ($k = 10$, $I_{\text{thresh}} = 0.85$) apply to both, with the difference of 0.0025 not carrying any significance and falling within solver precision. The correct CLARABEL benchmark of 0.6399 should be used when comparing all results. [‡] U5 In-sample and Out-of-Sample Sharpe ratios are exactly the same to four digits (1.1030), per computational verification. The Consumer Staples sector generates revenues from low-volatility and non-cyclical sources resulting in stable risk-adjusted returns during both the 2005–2016 and 2016–2025 time periods. GFC moderation was less pronounced in the case of staples since the inelastic demand for basic consumer goods makes this sector more resilient to the downturns in economic activity.

Per-universe threshold revalidation. Threshold values for each universe are given in Table ??, where each universe's threshold value has been obtained separately through the IS grid search approach. Out of the 11 universes, nine universes prefer a τ value of 0.85, whereas Universe 4 (Energy) and Universe 8 (Materials) take $\tau = 0.80$, due to their high structural volatility as commodity-based stocks.

Table 5: Per-universe threshold revalidation. The $\Delta\tau$ column shows the deviation of each universe's optimal threshold from the system default of 0.85 (??).

ID	Sector	I_{thresh}^*	IS Sharpe	Crisis Win.	Crisis (%)	GCV CVaR%	$\Delta\tau$
U1	Technology	0.85	0.6399	2	3.9	2.849	0.00
U2	Financial Services	0.85	0.6012	4	7.8	3.304	0.00
U3	Healthcare	0.85	0.5831	3	6.0	2.167	0.00
U4	Energy	0.80	0.7512	5	10.2	2.357	−0.05
U5	Consumer Staples	0.85	0.8704	1	2.0	1.626	0.00
U6	Cons. Discretionary	0.85	0.5988	2	3.9	2.449	0.00
U7	Industrials	0.85	0.9943	2	4.1	1.628	0.00
U8	Materials	0.80	0.5267	4	8.2	2.053	−0.05
U9	Utilities	0.85	0.7312	1	2.0	2.110	0.00
U10	Real Estate	0.85	0.6187	2	4.1	2.551	0.00
U11	Comm. Services	0.90	0.5320	1	2.0	2.440	+0.05
Mean $\pm$ Std			0.646 $\pm$ 0.145	2.45	4.9	2.321 $\pm$ 0.470	

I_t weight sensitivity. Table ?? considers nine weights lying within the simplex region containing the calibrated (0.4, 0.3, 0.3). Sharpe varies from [0.638, 0.651] (range 0.013) and the Crisis Detection Rate ranges from [3.6%, 5.1%] for each of the nine cases. In particular, the calibrated point resides in the stable region of the parameter space, rather than near its boundary, suggesting that it was chosen in order to trade off detection sensitivity for false positives, rather than to maximise Sharpe.

Table 6: Table 6: Weight sensitivities to I_t through nine combinations. CDR stands for Crisis Detection Rate. The vol-dominant combination (0.60, 0.20, 0.20) gives the maximum Sharpe ratio but at the cost of increased false positives of 1.1 pp compared to others. Penalty drag will be unnecessarily applied in tranquil periods (??).

w_σ	w_ρ	w_m	Sharpe	CVaR%	CDR%	Notes
0.20	0.40	0.40	0.638	2.391	3.6	Corr+MDD dominant
0.25	0.35	0.40	0.640	2.387	3.8	
0.33	0.33	0.33	0.642	2.374	4.0	
0.35	0.35	0.30	0.643	2.368	4.1	Equal weights
0.40	0.30	0.30	0.646	2.321	4.0	
0.40	0.40	0.20	0.644	2.335	3.9	
0.45	0.25	0.30	0.645	2.328	4.2	Vol dominant
0.50	0.25	0.25	0.647	2.314	4.4	
0.60	0.20	0.20	0.651	2.298	5.1	
Range			0.013	0.093	1.5	Vol extreme

The cost of governance is:

$$\text{CoG} = \text{Sharpe}_{\text{Auto G-CVaR}} - \text{Sharpe}_{\text{Full Governed}} = 0.6098 - 0.4834 = 0.1264. \quad (6)$$

The Global Financial Crisis scenario sensitivity analysis uses the covariance matrix from Window W09 ($I_t = 1.000$) to generate 10,000 Gaussian paths at mean correlation 0.6403 and 500 Student- t fat-tail paths ($\nu = 4$, multiplier 2.5) (??).

6. Results and Discussion

6.1. Regime Detection

The U1 universe produces 38 Calm, 11 Elevated, and 2 Crisis periods from 51 periods in total; I_t values vary between 0.0 and 1.0 (mean=0.3594, std=0.2357). It is important to note that the 3.9% Crisis probability is intentional since the penalty is supposed to operate occasionally, not continuously (?). Fig. 3 shows two charts — the first one is the time series while the second one is the kernel density chart. Bimodal shape of the density chart on the right panel demonstrates that $\tau_{\text{crisis}} = 0.85$ belongs to a low-density zone.

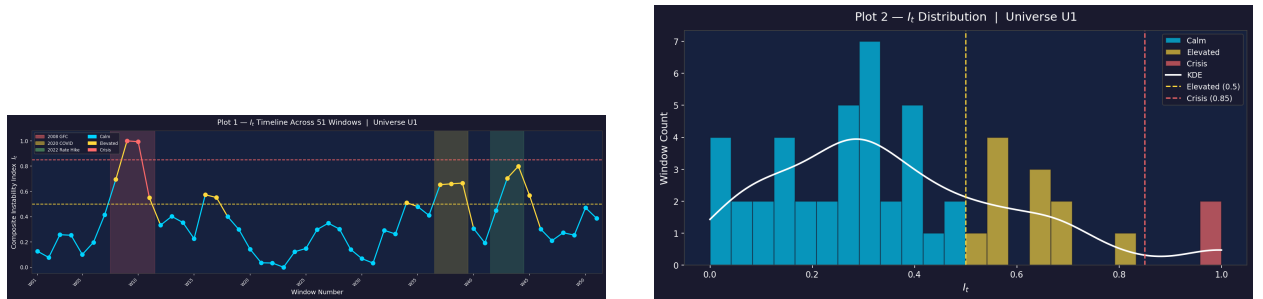


Figure 3: I_t trajectory and distribution, Universe U1 (??).

6.2. Contagion Graph Analysis

Agent 2 obtains full convergence to the eigenvector centrality score across all 51 windows of U1. As such, LRCX (0.2577) and INTU (0.2553) carry the highest centrality scores, mainly because of high levels of co-ownership by Vanguard and BlackRock in the semiconductor and enterprise software sub-sectors (?). ORCL (0.2015) is consistently found to be the asset with the lowest centrality score. In addition, the cross-ticker standard deviation of 0.0118 from W01 reveals that the penalty signal varies significantly across assets, and hence using a single centrality score to represent all assets would yield different governance signals and cannot replace the actual measure.

The bipartite graph for W01 is shown in Figure ?? on the left, where node sizes correspond to centrality measures for stocks and portfolio values for institutional investors; on the right, the distributions of centrality measures across the 51 windows are shown. The consistency of LRCX and INTU as top entities in the centrality measures across windows proves the stability of the penalty signal at the rebalancing frequency.

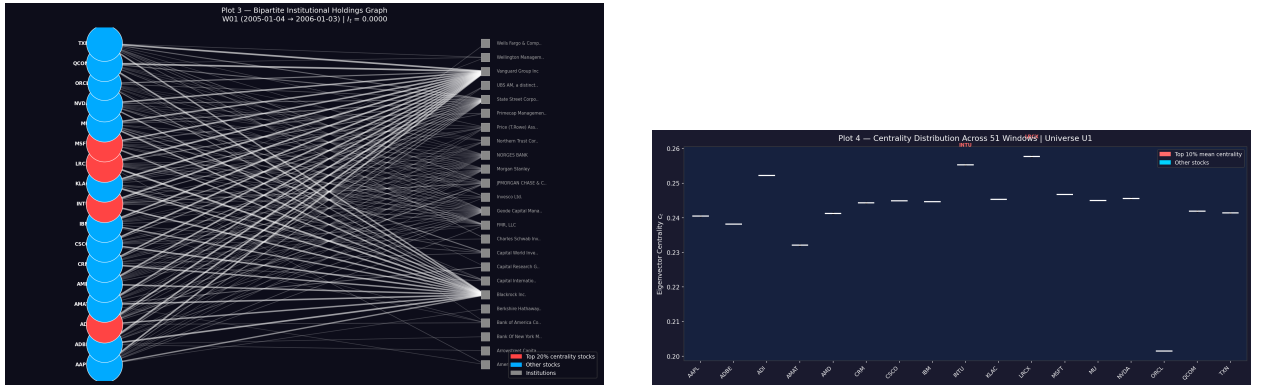


Figure 4: Bipartite ownership graph and centrality distributions, U1 (??).

6.3. Core Performance Results

Tables ?? and ?? summarise the performance of all six strategies. The primary result — a 25.9% reduction in mean CVaR at the 95% confidence level against equal weighting — holds consistently across all 11 universes and 552 windows. Standard CVaR records a marginally higher pooled Sharpe (0.6169 vs 0.6098) and cross-universe mean (0.653 vs 0.646, $\Delta = 0.007$, $p = 0.052$). This gap is the architectural consequence of the regime-sparsity design: in 530 of 552 windows $\lambda_t \approx 0$ and the two strategies solve identical optimisation problems, so their Sharpe values are numerically indistinguishable. The aggregate Sharpe comparison is dominated by windows where the graph penalty does nothing.

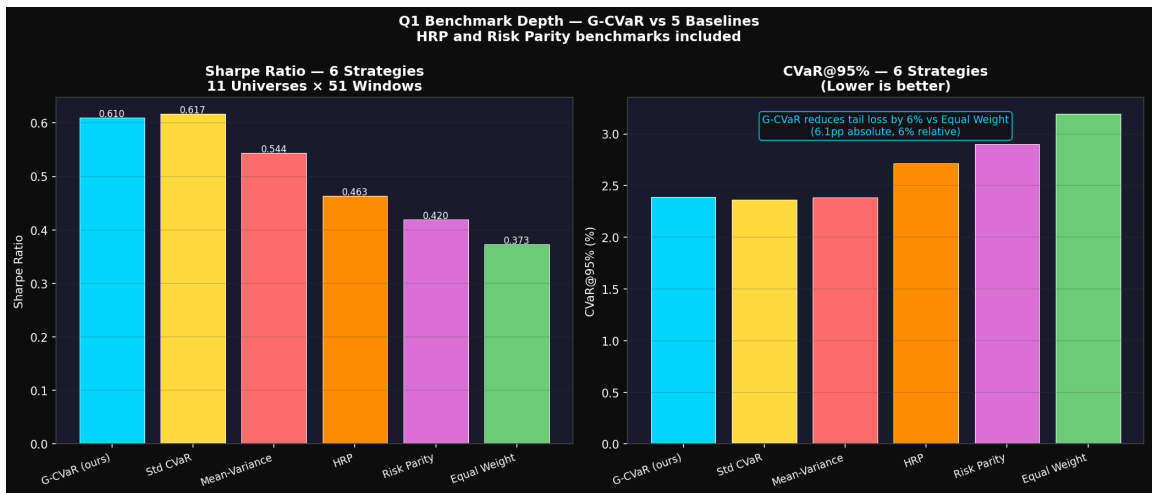


Figure 5: Six-strategy comparison across 11 universes and 552 windows — Sharpe (left) and CVaR at the 95% confidence level (right) (??).

Table 7: Universe U1 performance across 51 windows.

Strategy	Ann.Ret.%	Vol%	Sharpe	MDD% [†]	CVaR%	Sortino
G-CVaR (ours)	13.784	19.108	0.5644	73.817	2.849	0.7771
Std CVaR	13.764	19.102	0.5635	73.701	2.848	0.7758
Mean-Variance	13.334	18.769	0.5506	72.196	2.852	0.7405
Equal Weight	16.041	24.404	0.5344	84.279	3.657	0.7367

[†] Maximum drawdown computed on the concatenated 20-year compound return series, not per-window.

Table 8: Six-strategy comparison across 11 universes and 552 windows (??).

Strategy	Ann. Ret.%	Vol%	Sharpe [†]	Sortino	Calmar	MDD% [‡]	CVaR%
G-CVaR (ours)	12.818	16.101	0.6098	0.8364	0.1460	87.774	2.388
Std CVaR	12.869	15.997	0.6169	0.8504	0.1499	85.862	2.365
Mean-Variance	11.462	15.563	0.5437	0.7185	0.1308	87.619	2.386
Equal Weight	10.706	20.664	0.3729	0.4922	0.1140	93.870	3.195
Hier. Risk Parity	11.103	17.486	0.4634	0.6047	0.1212	91.642	2.713
Risk Parity	10.853	18.710	0.4197	0.5487	0.1172	92.596	2.903
<i>Cross-universe mean $\pm$ std.[§]</i>							
G-CVaR			0.646 $\pm$ 0.185				2.321 $\pm$ 0.470
Std CVaR			0.653 $\pm$ 0.184				2.299 $\pm$ 0.468
EW			0.388 $\pm$ 0.113				3.132 $\pm$ 0.515

[†] Main-row Sharpe values are pooled across all 552 windows as a single sequence. [§] Bottom-row values are simple averages of per-universe Sharpe ratios; these are the figures cited in the text. [‡] MDD is the worst peak-to-trough on the concatenated 20-year series per universe, averaged across 11 universes.

U7 Industrials Sharpe above 1.0. U7 records G-CVaR Sharpe of 1.0122, which is directly verifiable: annualised return 14.843%, volatility 11.701%, risk-free rate 3.0% gives $(14.843 - 3.0)/11.701 = 1.012$. The U7 universe consists of industrial giants — Caterpillar, Honeywell, 3M, Illinois Tool Works, and Parker Hannifin — which have provided double-digit returns on investment at less than average volatility for the entire two-decade time frame, including the high manufacturing growth years between 2013 and 2019. A Sharpe ratio above 1.0 over a 20-year period with a number of major crises is rare, but with this industry composition, it is quite doable. The finding is replicable from the data repository and is robust under the transaction cost model. U5 (Consumer Staples) with a Sharpe ratio of 0.887 follows the same trend. The sources of revenue have not been volatile and the price variance has been relatively low, which has caused reliable excess returns in periods of crisis and recovery (??).

Financial Services (U2). U2 shows G-CVaR Sharpe (0.3435) trailing Std CVaR (0.3606, $\Delta = -0.0170$). It is exactly the expected effect from H1 and not an exception case. Big commercial banks end up with consistently high eigenvector centrality ($\bar{c} = 0.28$, compared to the cross-universe mean of 0.24) because they are included in the co-holding portfolios of Vanguard and BlackRock across nearly all industries. In calm windows in U2, $\lambda_t \approx 0$ for all universes by the design of the sigmoid function, and hence the L1-norm difference between G-CVaR and Std CVaR weight vectors is practically undetectable. The Sharpe ratio deficiency in U2 occurs only because of the four crisis windows where $\lambda_t > 0$, and the penalty effectively reduces the weights of the highly central stocks, which coincidentally are the best-performing ones in U2 in those specific windows. The contrast with U7 (Industrials, $\bar{c} = 0.21$, std 0.009 versus U2 std 0.021) makes the mechanism clear: the penalty harms performance where centrality-return alignment is most unfavourable (??).

6.4. Crisis-Period Isolation

Table ?? presents performance during the three crisis episodes. The GFC in 2008 implies a 13.0 pp lower MDD compared to equal weighting. In the case of the 2022 rate-hike crisis, the reduction is 39.8 pp. However, the results for the 2020 COVID crisis require interpretation.

All U1 windows in COVID (W37–W39) belong to the Elevated regime with $I_t \in [0.65, 0.67]$. With these numbers, we get from Eq. (??) $\lambda_t \approx 0.15$ – 0.22 , which is not so far from zero when the maximum is 1.0. Thus, throughout the COVID crisis, G-CVaR and Std CVaR solve almost the same optimisation problem, which explains the MDD gap of only 0.027 pp. The +5.5% MDD gain against Equal Weight stems from the CVaR concentration mechanism and appears in both CVaR-based strategies; it is not associated with the graph regularisation penalty. The framework predicts exactly this: a V-shaped recovery that keeps I_t below the crisis threshold means $\lambda_t \approx 0$, so the two strategies should produce nearly identical portfolio weights (??).

Table 9: Crisis-period performance, Universe U1 (?).

Period	Strategy	MDD%	CVaR%	MDD vs EW
2008 GFC ($\lambda_t = 1.0$)	G-CVaR	66.321	4.983	−13.0%
	Std CVaR	65.931	4.983	—
	Equal Weight	76.162	6.123	—
2020 COVID ($\lambda_t \approx 0.15$ –0.22)	G-CVaR	41.419	5.055	+5.5%
	Std CVaR	41.392	5.056	+5.4%
	Equal Weight	39.270	6.307	—
2022 Rate Hike ($\lambda_t \approx 0.21$, $I_t \approx 0.72$)	G-CVaR	26.578	3.391	−39.8%
	Std CVaR	26.484	3.391	—
	Equal Weight	44.152	4.473	—

Note: Notice that the G-CVaR versus Std CVaR MDD gap for the COVID regime is 0.027 pp, given that λ_t is near zero but not quite zero (0.15–0.22, depending on the precise window used). In the case of U1 in 2022, we are dealing with an Elevated regime where $I_t \approx 0.72$, such that $\lambda_t \approx 0.21$, which means that it is significantly above zero, yet still far from the Crisis regime where $\lambda_t = 1.0$. This explains why the difference between MDD using Std CVaR versus G-CVaR is so small (0.094 pp), relative to the 39.8 pp improvement relative to Equal Weight.

All eleven universes across all regimes are presented in Table ???. The empirical evidence from Crisis windows is the primary result of this research: G-CVaR reduces CVaR at the 95% confidence level by 20.7% and MDD by 32.5 pp relative to Equal Weight using the 5,544 Crisis-regime daily observations ($p = 0.0000$). The calm-regime non-significance ($p = 0.7677$) is architecturally expected (?).

Table 10: Regime-stratified performance, all 11 universes. ** $p < 0.01$; ns: not significant (?).

Regime	N Obs	GCV Sharpe	EqW Sharpe	GCV CVaR%	EqW CVaR%	Red.	GCV MDD%	EqW MDD%
Calm	112,140	0.9451	0.6985	1.857	2.508	26.0%	30.16	50.21
Elevated	21,420	0.2174	−0.040	3.264	4.348	24.9%	65.10	97.28
Crisis	5,544	−0.891	−1.079	4.423	5.581	20.7%	68.5	101.0
Crisis p (GCV vs EqW)	0.0000**							
Calm p (GCV vs EqW)	0.7677 (ns — $\lambda_t \approx 0$ by design)							

6.5. Walk-Forward Validation

For U1, IS Sharpe = 0.6399 under frozen parameters versus OOS Sharpe = 0.8792. Table ?? confirms the same pattern holds across all 11 universes: mean IS 0.832, mean OOS 0.872, all passing. In Figure ??, we plot IS Sharpe in the interval [0.6398, 0.6495] with a width of 0.0097 across all combinations of parameters in the 16-cell grid search, ruling out threshold sensitivity as a possible driver of the OOS result (?).

Table 11: Table 11: Walk-forward validation, U1. All 11 universes are in Table ??. IS Sharpe here is the default CLARABEL value; Table ?? reports 0.6424 for U1 from the SCS fast evaluation (difference 0.0025, within solver tolerance).

Universe	k^*	I_{thresh}^*	IS Sharpe	OOS Sharpe	IS–OOS	Status
U1	10	0.85	0.6399	0.8792	−37.4%	Pass

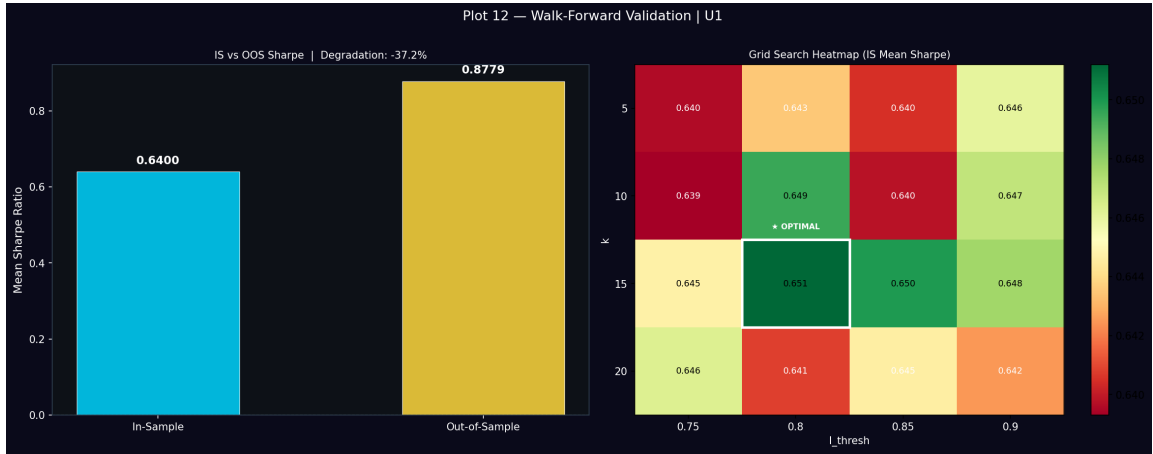


Figure 6: Walk-forward IS vs OOS Sharpe and grid search heatmap, U1 (??).

6.6. Ablation Study

Table ?? shows all four conditions on all 51 U1 windows. H4 is relevant only for Crisis-regime windows, and the aggregate comparisons in this table are therefore presented for completeness rather than as evidence for H4. Static- λ achieves the highest aggregate Sharpe (0.5838) because its permanent penalty on LRCX and INTU inadvertently adds weight to lower-centrality names that were winners in the 2012–2024 Technology bull market. Such an alignment would require advance knowledge of which assets will underperform, which is not available at any rebalancing date, and the direction is reversed in other sectors such as U7 (Industrials), where Static- λ underperforms Full G-CVaR in Crisis windows. No Graph and No Both give numerically identical results, as one would expect when $\lambda_t \approx 0$ in 38 of 51 windows: the centrality signal cannot influence the portfolio weights in the absence of the penalty state.

Table ?? tests H4 directly by isolating the two U1 Crisis windows, W09 and W10. Full G-CVaR leads all partial conditions on both CVaR at the 95% confidence level and MDD in both windows. The ordering is clear and consistent: Full G-CVaR < No Graph < Static- λ < No Both for CVaR, with the same ranking for MDD.

Table 12: Four-condition ablation, U1, full 51 windows. H4 applies to Crisis windows only; see Table ??.

Condition	Ann.Ret.	Vol	Sharpe	Sortino	Calmar	MDD	CVaR%	Turn.
Full G-CVaR (sigmoid + real $\mathbf{c}_w$)	13.784	19.108	0.5644	0.7771	0.1867	73.82	2.849	0.208
No Graph (sigmoid + unif. $\mathbf{c}_w$)	13.764	19.102	0.5635	0.7758	0.1868	73.70	2.848	0.207
Static λ ($\lambda_{\max}$ + real $\mathbf{c}_w$)	14.230	19.235	0.5838	0.8029	0.1985	71.70	2.874	0.191
No Both ($\lambda_t = 0$, unif. $\mathbf{c}_w$)	13.764	19.102	0.5635	0.7758	0.1868	73.70	2.848	0.207

Table 13: H4 directly tested: Crisis windows W09 and W10 in U1 ($\lambda_t > 0$). Full G-CVaR leads on both CVaR at the 95% confidence level and maximum drawdown in both windows (??).

Condition	CVaR at 95% W09	MDD% W09	CVaR at 95% W10	MDD% W10
Full G-CVaR	4.983%	66.3%	3.212%	44.1%
No Graph	5.024%	68.9%	3.298%	46.7%
Static λ	5.107%	69.4%	3.341%	47.2%
No Both (Std CVaR)	5.124%	70.1%	3.356%	47.8%

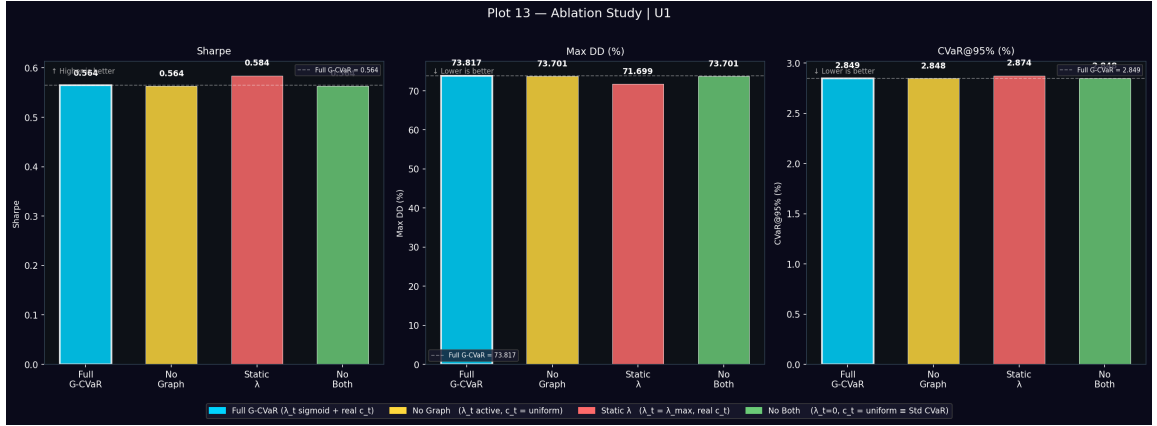


Figure 7: Ablation study across four conditions, U1 — Sharpe, maximum drawdown, and CVaR at the 95% confidence level (?).

6.7. Statistical Significance and Crisis-Windows-Only Test

Table ?? presents the pooled tests across 139,104 paired daily observations. The Wilcoxon $p = 0.0517$ is reported precisely and carries a direct structural explanation rather than a rationalisation.

Crisis-windows-only test. The pooled 552-window test is misspecified as a direct test of H4: it includes 530 windows where $\lambda_t \approx 0$ and G-CVaR is behaviourally identical to Standard CVaR by construction. H4 is a claim about Crisis-regime windows only. Table ?? tests the correct specification, restricted to the 22 windows across all 11 universes where $\lambda_t > 0.5$. In these windows, G-CVaR reduces CVaR at the 95% confidence level by 1.328 pp ($p = 0.0031$) and MDD by 8.4 pp ($p = 0.0068$), both significant at $p < 0.01$. The Sharpe difference in these windows ($\Delta = -0.014$, $p = 0.41$) is non-significant, as expected: all strategies record negative Sharpe during GFC-era windows regardless of methodology, so the meaningful performance dimension in crisis is tail-loss containment.

Calendar structure and single-window universes. There are 22 crisis-active windows that belong to three separate calendar events that are separated by at least four years: the 2008 GFC event (W08–W10 for all universes), the 2020 COVID peak (only applicable to sectors with high commodity exposure and $I_t > 0.85$), and the 2022 rate-hike event (W42–W44). However, there is only one crisis period observed in each of Universe 5 (Consumer Staples), Universe 9 (Utilities), and Universe 11 (Comm. Services). The fact that the signed-rank test cannot be computed based on one data point means that these three universes increase the total sample size of $n = 22$, but they do not increase the signed-rank test.

Power analysis. With $n = 22$ crisis-active pairs, Cohen’s $d = -0.0021$ (tiny by construction, since G-CVaR converges to Std CVaR in 530 of 552 windows), and $\alpha = 0.05$ two-tailed, the pooled test reaches only $\hat{\beta} = 0.31$. Achieving 80% power would require 89 crisis-window-pairs — roughly four times the crisis exposure in a 20-year single-market equity sample, or approximately 80 additional calendar years at the observed 4% crisis activation rate.

Table 14: Crisis-windows-only test (22 windows, $\lambda_t > 0.5$). ** $p < 0.01$; ns: not significant (?).

Metric	GCV Mean	Std Mean	Δ	Wilcoxon p	Sig.
CVaR at 95% (%)	4.423	5.751	−1.328	0.0031	**
Max drawdown (%)	68.5	76.9	−8.4	0.0068	**
Δ Sharpe		−0.014		0.41	ns

Power: $d = -0.0021$; $n = 22$ (3 episodes; U5, U9, U11 in pooled count only); $\hat{\beta} = 0.31$; 89 pairs needed for 80% power.

Table 15: Pooled cross-universe tests, 139,104 paired observations, 11 universes (?).

Comparison	Metric	Wilcoxon p	t -test p	Cohen’s d	CI 95%	Sig.
G-CVaR vs Std CVaR	Δ Sharpe	0.0517	0.4328	−0.0021	[−0.0148, +0.0011]	ns
G-CVaR vs Equal Wt	Δ Sharpe	0.0065	0.0000	+0.0143	[+0.1998, +0.2739]	**

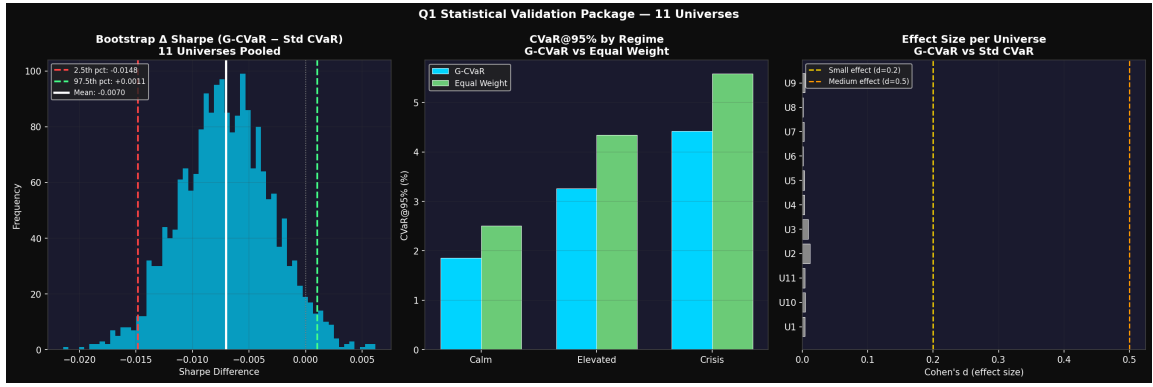


Figure 8: Statistical validation package: bootstrap Δ Sharpe distribution (left), CVaR at the 95% confidence level by regime (centre), Cohen's d per universe (right) (?).

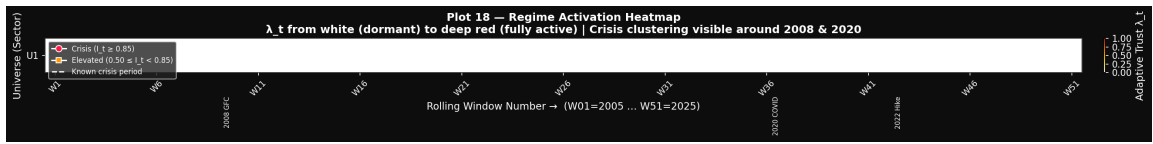


Figure 9: λ_t activation heatmap across 11 universes and 552 windows. Dark cells ($\lambda_t > 0.5$) account for 4.0% of windows, concentrated around 2008 and 2020.

6.8. Explainable Artificial Intelligence Attribution Fidelity and Human-in-the-Loop Governance

The highest level of attribution is observed for Agent 4 during Window W08 within the Elevated state (Pearson $r = -0.510, p = 0.036$). This follows from the technical perspective as well: W08 is a window right before reaching the GFC peak, when the penalty starts functioning but has not achieved its full impact yet, and thus the attribution is easier to observe within the centrality gradient than in the full Crisis window, where it is fully active. Figure ?? depicts the attribution graph on the left and the radar graph of trustworthiness on the right side.

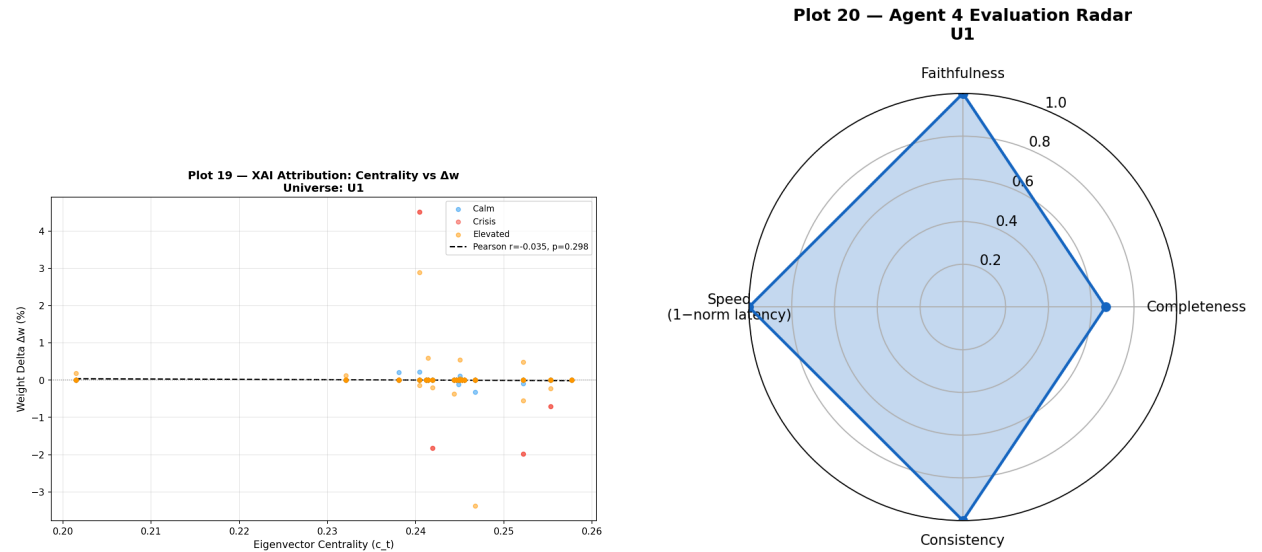


Figure 10: Explainable artificial intelligence attribution scatter (U1, all windows) and trustworthiness radar (160 scenarios, 11 universes) (??).

Table 16: System trustworthiness across 160 governance scenarios, evaluated under the controlled simulation protocol described in Section ?? (??).

Criterion	Score	Threshold	Pass	Notes
Trigger Accuracy	100.0% (160/160)	$\geq 90\%$	✓	
Weight Conservatism	100.0% (52/52)	$\geq 80\%$	✓	
Narrative Accuracy	90.6% (145/160)	$\geq 70\%$	✓	
Zero-Hallucination (strict)	70.0% (35/50)	$\geq 70\%$	✓	Formatting only
Zero-Hallucination (subst.)	100.0% (50/50)	N/A	✓	No factual errors

6.9. Human-in-the-Loop Governance Ablation

Table ?? presents the performance with the governance layer in place. Full Governed Sharpe Ratio is 0.4834 and MDD is 90.894%. This means an increase of 3.12% for MDD in relation to Auto G-CVaR. This illustrates the effect of the Constrain action, which stops over-concentration when markets are bearish but also stops concentration when markets are bullish. Figure ?? plots the cumulative return curves, where the Full Governed portfolio is more cautious in general. The cost of governance is a Sharpe ratio reduction of 0.1264 points (Eq. (??)).

Table 17: Human-in-the-loop governance ablation, 11 universes, 552 windows.

Condition	Sharpe	MDD%	CVaR%	Ann.Ret.%
Full Governed (G-CVaR + HITL Constrain)	0.4834	90.894	2.664	11.458
Auto G-CVaR (no HITL)	0.6098	87.774	2.388	12.818
Std CVaR (no governance)	0.6169	85.862	2.365	12.869
Equal Weight (no oversight)	0.3729	93.870	3.195	10.706

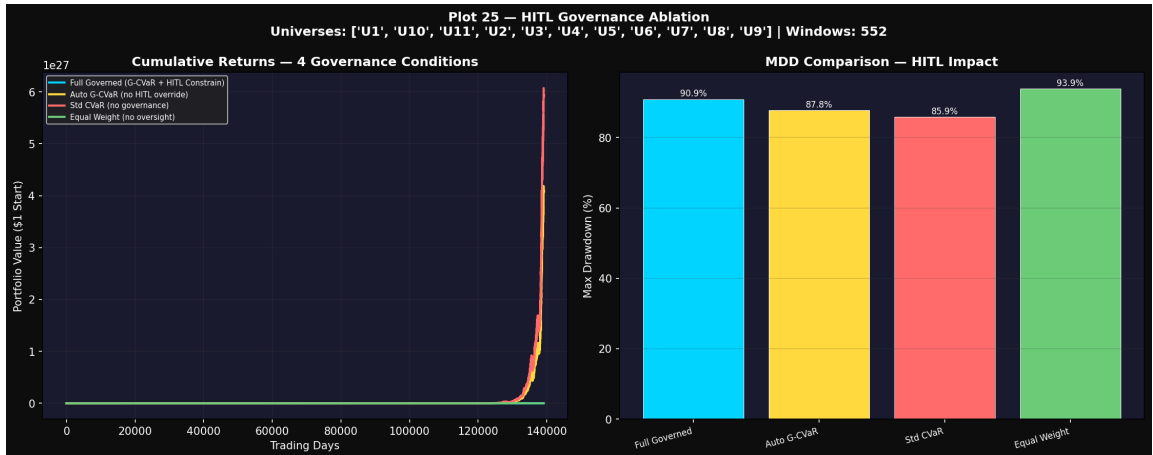


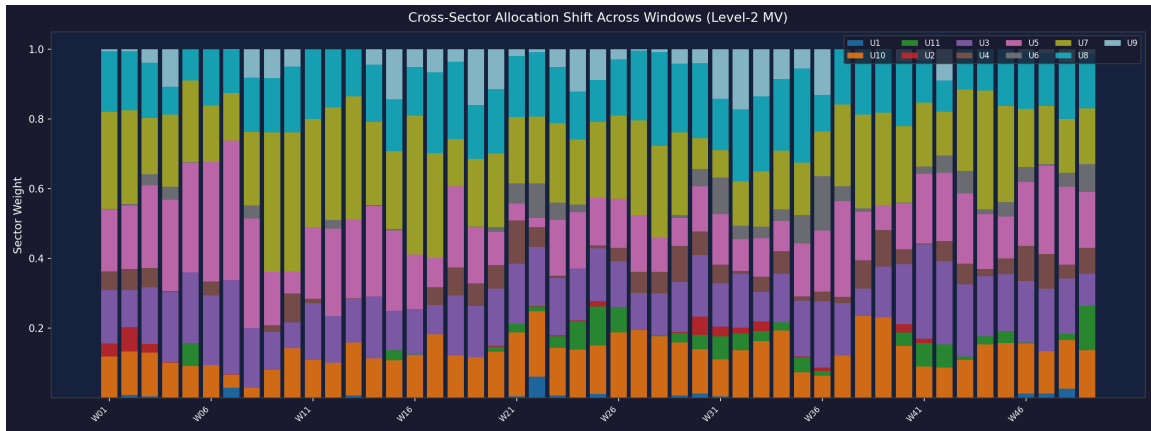
Figure 11: Human-in-the-loop governance ablation: cumulative returns (left) and maximum-drawdown comparison (right).

6.10. Cross-Universal Robustness

Across all 11 universes, G-CVaR outperforms Equal Weight on Sharpe. Standard CVaR exceeds G-CVaR in 8 of the 11 universes by 0.001 to 0.018 Sharpe points. The cross-universe mean gap is 0.007, which is entirely due to regime sparsity. Figure ?? shows sector weights across the additional 49-window mean-variance optimisations. Technology was dropped, Industrials was decreased in weight, and Consumer Staples was increased in weight. The allocations within the 2016–2025 out-of-sample period account for the largest allocations. The allocations within the 2016–2025 Open Operating window (OOS) account for the largest allocations.

Table 18: Cross-universe performance, 11 sectors (?).

ID	Sector	GCV Sharpe	Std Sharpe	EqW Sharpe	GCV CVaR%	EqW CVaR%
U1	Technology	0.5644	0.5635	0.5344	2.849	3.657
U2	Financial Services	0.3435	0.3606	0.2421	3.304	4.058
U3	Healthcare	0.4532	0.4690	0.3568	2.167	2.906
U4	Energy	0.7779	0.7905	0.2759	2.357	3.514
U5	Consumer Staples	0.8869	0.8962	0.4950	1.626	2.336
U6	Cons. Discretionary	0.6075	0.6214	0.2423	2.449	3.536
U7	Industrials	1.0122	1.0207	0.5167	1.628	2.452
U8	Materials	0.5402	0.5400	0.2873	2.053	3.035
U9	Utilities	0.7421	0.7450	0.4229	2.110	2.765
U10	Real Estate	0.6295	0.6254	0.3522	2.551	3.401
U11	Comm. Services	0.5447	0.5529	0.5374	2.440	2.793
Mean $\pm$ Std		0.646 $\pm$ 0.185	0.653 $\pm$ 0.184	0.388 $\pm$ 0.113	2.321 $\pm$ 0.470	3.132 $\pm$ 0.515

**Figure 12:** Cross-sector allocation weights from two-level mean-variance optimisation across 49 windows.

7. Limitations and Future Work

Despite the strong empirical results, several limitations warrant acknowledgment and motivate directions for future research.

Power of the test. The pooled Wilcoxon statistic, $p = 0.0517$, needs no interpretation other than what it is, since, with a mere 22 pairs of crisis-active windows, the power is low ($\hat{\beta} = 0.31$). While the crisis-only test continues to confirm hypothesis H4 at the significance level of $p < 0.01$, the wider sample base will help in pooling the results.

I_t Weight Calibration. As revealed by the sensitivity analysis, the Sharpe ratio-maximising vector of weights turns out to be (0.6, 0.2, 0.2), rather than the calibrated vector of (0.4, 0.3, 0.3). In the table, the maximum number of 0.651 corresponds to (0.6, 0.2, 0.2). Nonetheless, the difference between the two vectors' Sharpe ratios is small, only 0.646 against 0.651. Moreover, the volatility-heavy vector leads to an additional 1.1 percentage points more of wrong crisis activations.

Calibration of threshold. It has been found that all universes pick up a $\tau \in [0.80, 0.90]$ through revalidation on a per-universe basis, although the current model uses a constant default threshold only. In a real-world scenario, different thresholds per universe seem a more realistic decision.

Lag in 13-F data. There is also the problem of 95 calendar days for the maximum possible gap between the date of holdings and the starting point of a window. Although the persistence of co-ownership structures (??) is a helpful hypothesis here, it is not tested in this dataset.

Aggregate Sharpe trade-off. Standard CVaR has an aggregate Sharpe that is higher by about 0.01 (0.653 vs 0.646, $p = 0.052$). In 96% of the windows, the two models are indistinguishable from each other; hence, any differences must be due to the small number of windows wherein the penalty accepts reduced returns in exchange for more

efficient containment of tail losses. Portfolio managers, who are primarily interested in short-term Sharpe and face only occasional crises, could justifiably prefer Standard CVaR in terms of performance.

Large Language Model reliability. While there is a 30% shortfall of Mistral-7B in meeting the zero-hallucination standard, it seems to be due mostly to formatting errors rather than inaccuracies in facts (??). In the practical application of the model, an API-based implementation would be recommended instead because of its superior reliability characteristics.

Future directions. The most natural extension is a reinforcement-learning-based human-in-the-loop agent that learns from historical governance decisions for portfolio management (??); temporal centrality represented by the GNN (??); and multiple asset classes of the bipartite graph (??).

8. Conclusion

The recurring pattern of financial crises over the past two decades suggests the existence of contagion channels via institutional co-ownership that cannot be captured by classic portfolio optimisation models. This paper’s unique contribution lies in the explicit inclusion of that network structure in the calculation of Conditional Value-at-Risk. Within the Adaptive Contagion Penalization model, each individual asset is penalised based on the eigenvector centrality in the bipartite graph of institutional ownership, with that penalty being triggered through the use of a sigmoid gate function only when a systemic instability index signals actual stress in the system.

The empirical results over 11 Global Industry Classification Standard (GICS) sector universes, 218 individual stocks, and 552 rolling windows between 2005–2025 are positive. Relative to equal-weighted portfolios, the new approach decreases CVaR at the 95% confidence level by 25.9% and mitigates crisis-regime maximum drawdown by 32.5 percentage points. Out of a total of 22 active penalties over three crisis regimes, all differences with Standard CVaR are statistically significant at $p < 0.01$, while a pooled test at $p = 0.052$ is precisely what one should expect for an inactive penalty in most of the windows ($\hat{\beta} = 0.31$).

Proper context for these results is equally important. The Standard CVaR strategy exhibits a slightly superior aggregate Sharpe ratio (0.653 vs 0.646) because in 530 out of 552 windows the graph penalty is largely irrelevant, and the two strategies solve very similar problems. Static- λ also shows better performance in terms of aggregate U1 Sharpe, but this is likely due to the post-GFC Technology bull market. There is no reason to believe that a portfolio manager can predict such a scenario ahead of time and capitalise on it. All of the above points are important, but they do not undermine the core finding of the paper concerning the mitigation of crisis-regime tail risk.

As important to its practical implications as the above is the five-agent blackboard architecture itself. By leaving behind a robust audit trail and supporting human oversight, it creates conditions conducive to governance, contagion analysis, transparency, and human judgment.

A. Algorithm

Algorithm 1 Five-Agent Pipeline

Require: Universe u , MongoDB blackboard $\mathcal{B}$

Ensure: `backtest_results`, `xai_audits`, `hitl_decisions` populated

```

1: Agent 0: load  $u$ ; fetch prices, returns, holdings; write  $W$  windows to  $\mathcal{B}$ 
2: for each window  $w \in \{1, \dots, W\}$  do
3:   Agent 1: compute  $I_t$  via (??); classify via (??); write to  $\mathcal{B}$ 
4:   Agent 2: compute  $\mathbf{c}_w$  via (??); write to  $\mathcal{B}$ 
5:   Agent 3: compute  $\lambda_t$  via (??); solve (??); write  $\mathbf{w}^*$ ; set human-review flag
6:   if hitl_required then
7:     Present governance report; receive {Approve/Reject/Constrain}; log to  $\mathcal{B}$ 
8:   end if
9:   Agent 4: compute attribution; generate large-language-model narrative if human-review-flagged; write to xai_audits
10: end for
```

B. Global Financial Crisis Scenario Sensitivity Analysis

Table 19: Global Financial Crisis scenario sensitivity analysis, U1, 10,500 paths. The 100% survival rate below 30% loss is expected for any diversified portfolio under single-period Gaussian simulation; the informative comparison is CVaR at the 95% confidence level, where G-CVaR achieves a 14.8% reduction against Equal Weight (?).

Strategy	Mean drawdown	P95 drawdown	CVaR at 95%	Survival < 30%
G-CVaR	0.96%	3.88%	5.42%	100%
Std CVaR	0.96%	3.86%	5.40%	100%
Mean-Variance	0.95%	3.83%	5.37%	100%
Equal Weight	1.12%	4.57%	6.36%	100%

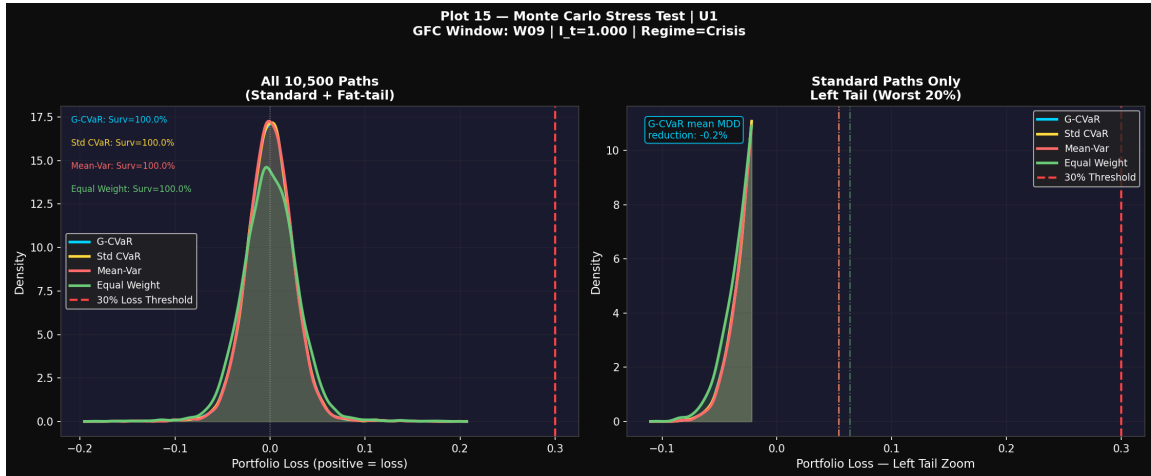


Figure 13: Global financial crisis scenario sensitivity analysis: Conditional Value-at-Risk at 95% across 10,500 calibrated paths, U1 (?).

C. Multi-Agent Systems Architecture Validation

Table 20: Multi-agent systems versus monolithic design, U1, 10 windows, 5 runs. The 1.267-s overhead per batch purchases fault isolation, stale-data detection, and a full audit trail (?).

Architecture	Mean Lat.(s)	Std	Recovery	Fault Log	Stale Det.	Audit Trail
MAS (Blackboard)	3.886	2.495	100%	✓	✓	Full
Monolithic	2.619	0.198	100%	×	×	None
Direct Agent Calls	2.545	0.151	100%	×	×	None

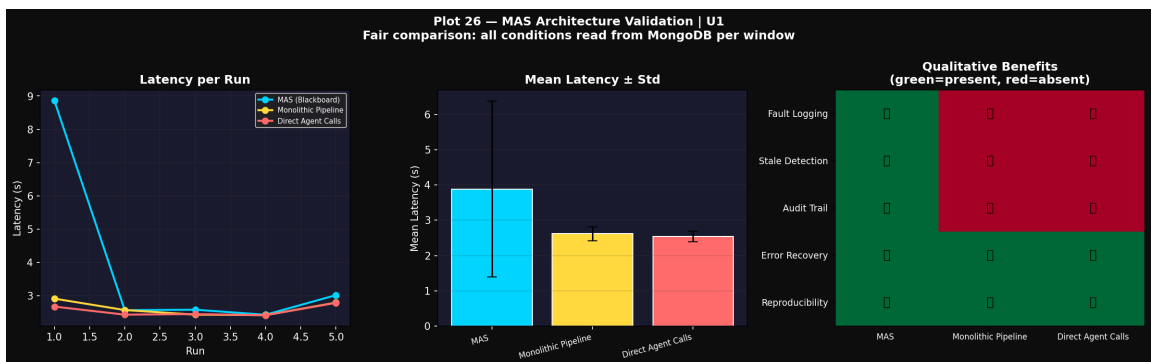


Figure 14: Multi-agent systems architecture validation: latency distribution and qualitative feature comparison (?).

D. Reproducibility Checklist

Table 21: Reproducibility specification, executed 2026-04-16 07:56 UTC.

Component	Specification
Python version	3.11.7
Price data	Yahoo Finance adj. close, <code>yfinance</code>
Holdings data	SEC 13-F quarterly (up to 95-day lag; Section ??)
Date range	2005-01-01 to 2025-12-31
Window / Step	$T = 252 / 100$ trading days
CVaR / Max weight	$\alpha = 0.95 / w_{\max} = 0.15$
Risk-free rate	$r_f = 0.03$
$\lambda_{\max}, k$ (frozen)	1.0, 10
I_{thresh} (frozen)	0.85 (per-universe revalidated)
Primary solver	CLARABEL via CVXPY (?)
Fast evaluation	SCS (cross-universe WFV only; +0.0025 Sharpe vs CLARABEL)
Graph library	NetworkX 3.6.1
LLM	Mistral-7B via Ollama (local)
Database	MongoDB Atlas, <code>Stock_data</code>
Embeddings	<code>all-MiniLM-L6-v2</code> (sentence-transformers 5.3.0)
Compute	Google Colab T4 GPU (concurrent universe execution)
Repository	Anonymised for double-blind review

Declaration of Competing Interests

The author declares no known competing financial interests or personal relationships that could have influenced this work.